REPUBLIC POLYTECHNIC

Lesson 4
Queuing Theory – Solution

E211 – Operations Planning II

Problem Statement

Figure 1: IKEA self-checkout area

Data gathered:

- Arrival rate = 150 customers per hour
- Mean checkout time = 2.5 minutes

Queuing Model

Appropriate queuing model for IKEA self-checkout is M/M/k. If number of self-checkout machines is 8, then the queuing model would be **M/M/8**.

Assumptions:

- Input source: Infinite population; no balks, no reneging
- Arrival process: Number of arrivals follows **Poisson distribution**
- Queue: Unlimited; single line
- Queue discipline: FIFO (FCFS)
- Service mechanism: **Multiple servers,** single phase system
- Service time: **Negative exponential distribution**
- Arrivals and service times are independent of the number of customers in system
- Arrival rate must be smaller than sum of service rate of servers, i.e.

  $\lambda < K\mu$ **(for K servers)**

Steady state equilibrium condition

Minimum Number of Servers, k

Arrival rate, λ = 150/60 = 2.5 per minute
Service rate, μ = 1/ 2.5 = 0.4 per minute

Steady state equilibrium holds when $\lambda < k\mu$ (for k servers)

Therefore, minimum k > λ/μ > 2.5/ 0.4 > 6.25

Minimum number of self-checkout machines, k = 7

Performance Measures

Arrival rate = 150 customers per hour
Mean checkout time = 2.5 minutes

Arrival rate	λ	2.500	per minute
Service rate	μ	0.4000	per minute
No. of servers	k	8	
Utilisation	ρ	0.7813	
Prob (no customers)	P_0	1.58E-03	
No. of customers in queue	Lq	1.4936	
No. of customers in system	Ls	7.7436	
Waiting time in queue	Wq	0.5975	minutes
Time in system	Ws	3.0975	minutes

- Utilization, ρ = 2.5/ (8 x 2.5) = 0.7813
- Average number of customers in queue, Lq = 1.4936
- Average number in system, Ls = Lq + λ/μ = 1.4936 + (2.5/ 0.4) = 7.7436
- Average waiting time Wq, = Lq/ λ = 1.4936/ 2.5 = 0.5975 min
- Average time in system Ws = Wq + $1/\mu$ = 0.5975 + (1/ 0.4) = 3.0975 mins

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Probability of at least 1 self-checkout available

No. of customers in system, n	$\frac{(\lambda/\mu)^n}{n!}$	Probability of n customers in system, Pn	
		$\frac{(\lambda/\mu)^n}{n!}P_0$ for $0 \le n \le k$	$\frac{(\lambda/\mu)^n}{k!k^{n-k}}P_0$ for $n > k$
0	1	1.584E-03	0.659221436
1	6.25	9.902E-03	0.515016747
2	19.53125	3.094E-02	0.402356834
3	40.690104	6.446E-02	0.314341276
4	63.578288	1.007E-01	0.245579122
5	79.47286	1.259E-01	0.191858689
6	82.784229	1.312E-01	0.149889601
7	73.91449	1.171E-01	0.117101251
8	57.745695	9.149E-02	0.091485352
9	40.101177	6.353E-02	0.071472931
10	25.063236	3.971E-02	0.055838228
11	14.240475	2.256E-02	0.043623615
12	7.416914	1.175E-02	0.034080949

Sum these values for $\sum_0^7 \frac{(\frac{\lambda}{\mu})^n}{n!} P_0$

No. of self-checkout machines, k = 8

Probability of at least 1 self-checkout machine is available,

= Probability of 7 or less customers in system

= P(n<=7) = $\sum_0^7 \frac{(\frac{\lambda}{\mu})^n}{n!} P_0$ = 0.582

REPUBLIC POLYTECHNIC

Probability of at least 2 customers in queue

No. of customers in system, n	$\frac{(\lambda/\mu)^n}{n!}$	Probability of n customers in system, Pn	
		$\frac{(\lambda/\mu)^n}{n!}P_0$ for $0 \le n \le k$	$\frac{(\lambda/\mu)^n}{k!k^{n-k}}P_0$ for $n > k$
0	1	1.584E-03	0.659221436
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12	7.416914	1.175E-02	0.034080949

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No. of self-checkout machines, k = 8

Sum these values for $\sum_0^9 \frac{(\frac{\lambda}{\mu})^n}{n!} P_0$

Probability of at least 2 customers waiting in the queue,

= Probability of 10 or more customers in system

= P(n>=10) = 1- $\sum_0^9 \frac{(\frac{\lambda}{\mu})^n}{n!} P_0$ = 0.255

Determining Service Time

Suppose that there is a cashier counter set up to assist customers who are unable to perform self-checkout. Your task is to determine an average service time standard for the cashier checkout so that the service level and staff utilization are reasonable. Initial estimates are that on average, 20 customers are expected to use the cashier checkout every hour.

Discuss the approach you would take to carry out your task.

Determining Service Time – M/G/1 model

Make use of M/G/1 model. Experiment with different values of Service Time to find out the corresponding service level.

M/G/1

Arrival Rate, λ	0.333	per minute
Service Rate, μ	0.400	per minute
Service Time, 1/μ	2.5	minute
Standard deviation of service times, σ	0.500	minute
System Utilization, ρ (set target)	0.833	
Probability of no customers, P_0	0.167	
Average number in queue, Lq	2.167	
Average time in queue, Wq	6.500	minutes
Average number in system, Ls	3.000	
Average time in system, Ws	9.000	minutes

Average service time can be regulated through adequate staff training and limiting the number of check out items for each customer.

Average service time impacts utilization and service level

Standard deviation may be adjusted with more data

Is utilization of 0.833 reasonable?

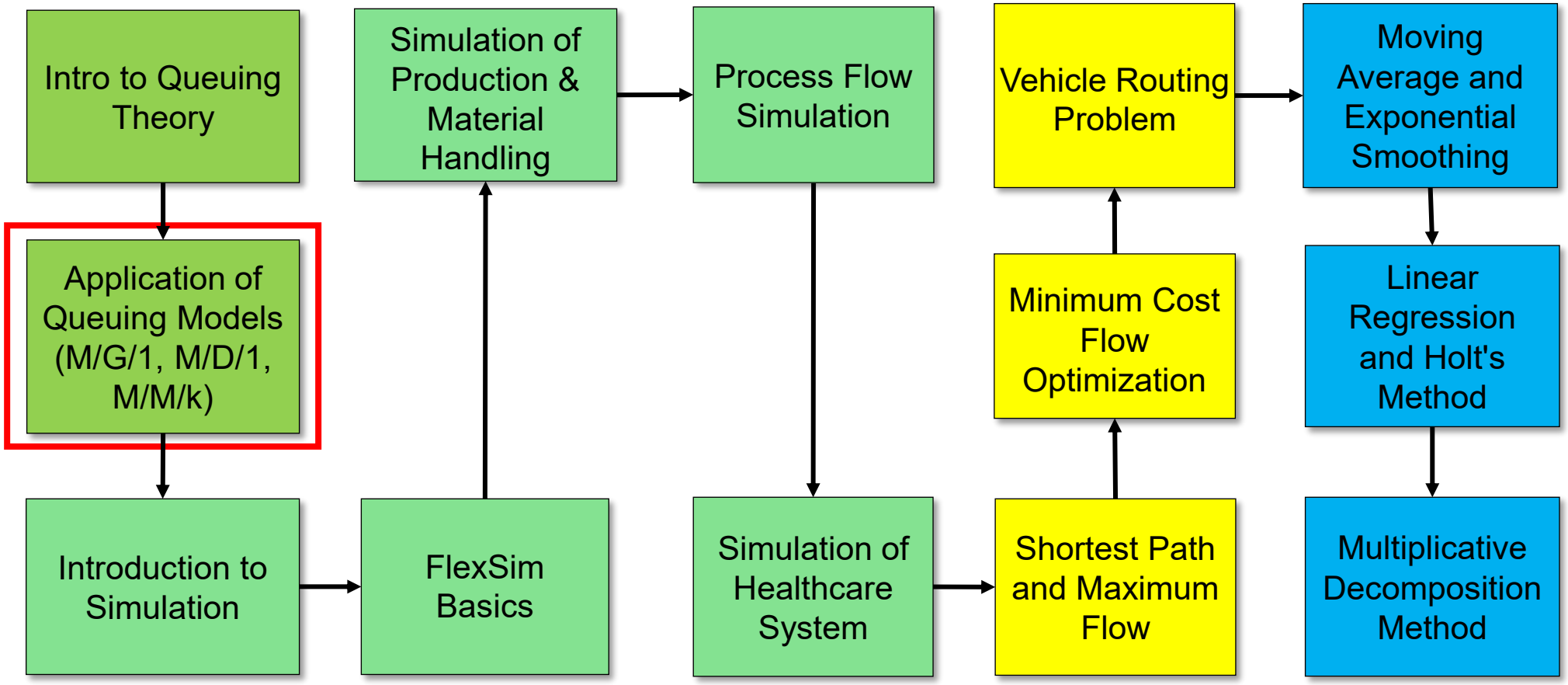
Are the average waiting time and time in system ok? Experiment with the service time to find a desirable service level.

Learning Outcomes

At the end of the lesson, students should be able to:

- Calculate performance measures of M/M/k queuing models.
- Build Excel models of queuing systems.
- Apply queuing models in the design of service operations.

E211 Progression



Appendix – Calculation of Lq

Average number in queue Lq = $\frac{\left(\frac{\lambda}{\mu}\right)^{k}\lambda\mu}{(k-1)!(k\mu-\lambda)\text{^}2}$ *P_0

$= \frac{(\frac{2.5}{0.4})^{8}*2.5*0.4}{(8-1)!(8*0.4-2.5)\text{^}2} * 0.158$

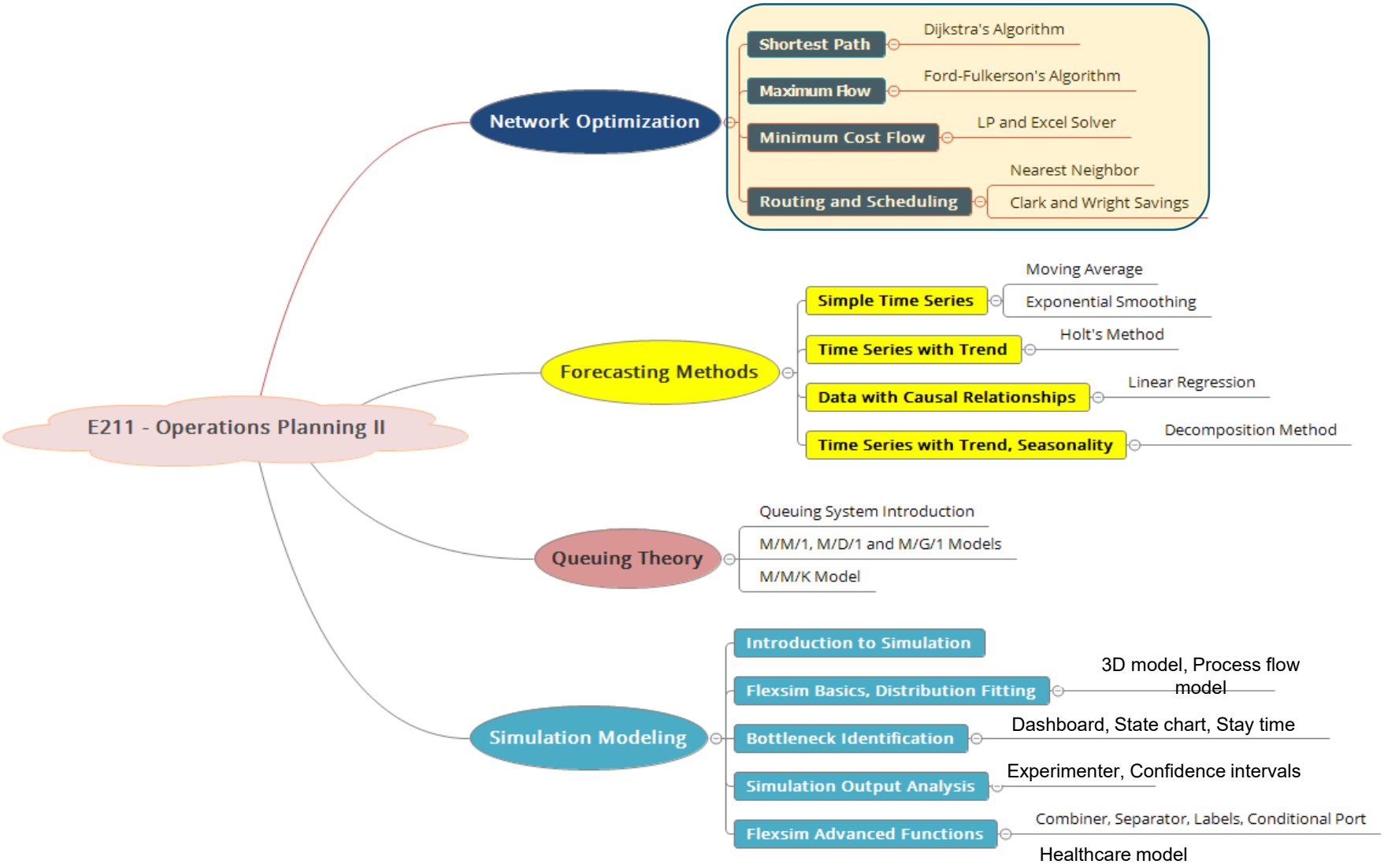
= 1.4936

P_0 is determined in the M/M/k template in Excel.

Lessons 16, 18 & 20 Solution

E211 – Operations Planning II

Module Coverage: E211 Topic Tree



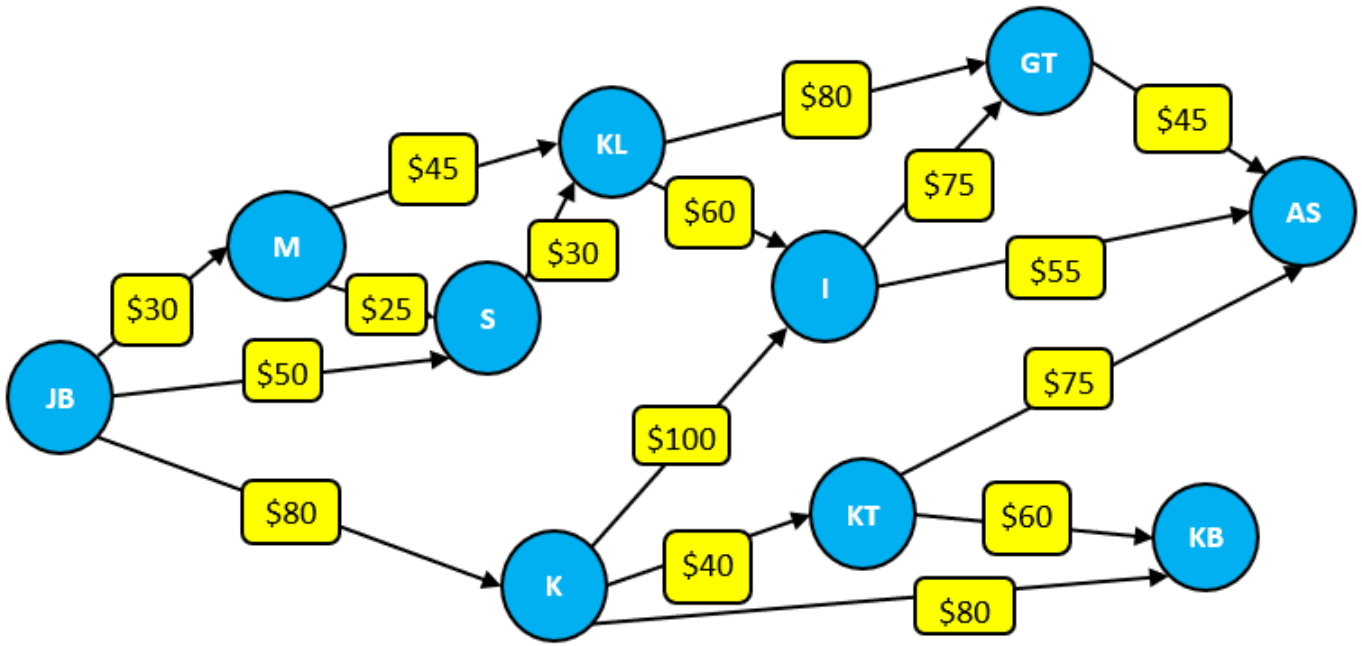
Lesson 16
Solution

Shortest path problem & Maximum flow problem

Learning Outcomes

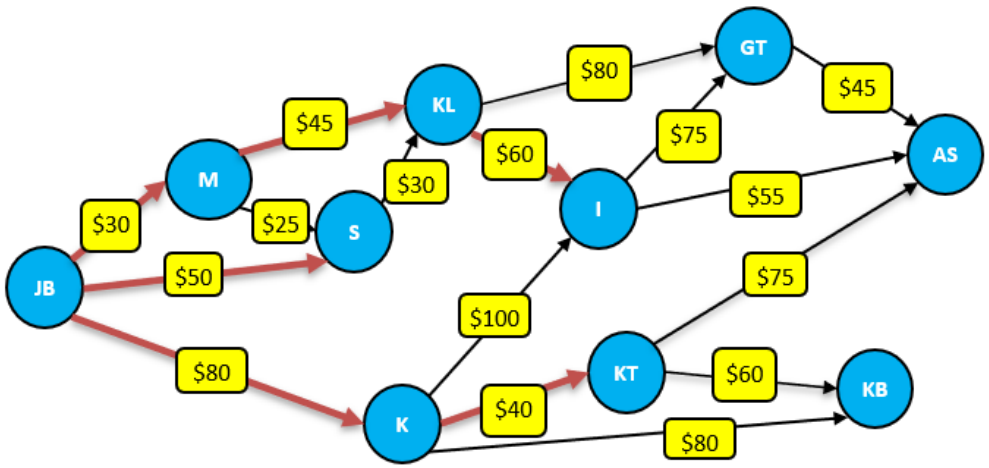
- Identify a Shortest Path problem.
- Apply Dijkstra's Algorithm to solve a Shortest Path problem.
- Identify a Maximum Flow problem.
- Apply Ford-Fulkerson's Algorithm to solve a Maximum Flow problem.

The Shortest Path Problem



- The delivery routes can be represented as a network model.
- The ten cities are represented by nodes.
- The delivery routes are represented by arcs.
- Minimum cost delivery route from Johor Bahru to each destination can be determined by solving as a shortest path problem either using the Dijkstra's Algorithm or LP model.

Dijkstra's Algorithm



Iteration	Solution Set	Arcs	Total Cost
Iteration 1	JB	**JB - M** JB - S JB - K	**30** 50 80

Iteration	Solution Set	Arcs	Total Cost
Iteration 2	JB M (30)	**JB - S** JB - K M - S M - KL	**50** 80 30+25 = 55 30+45 = 75

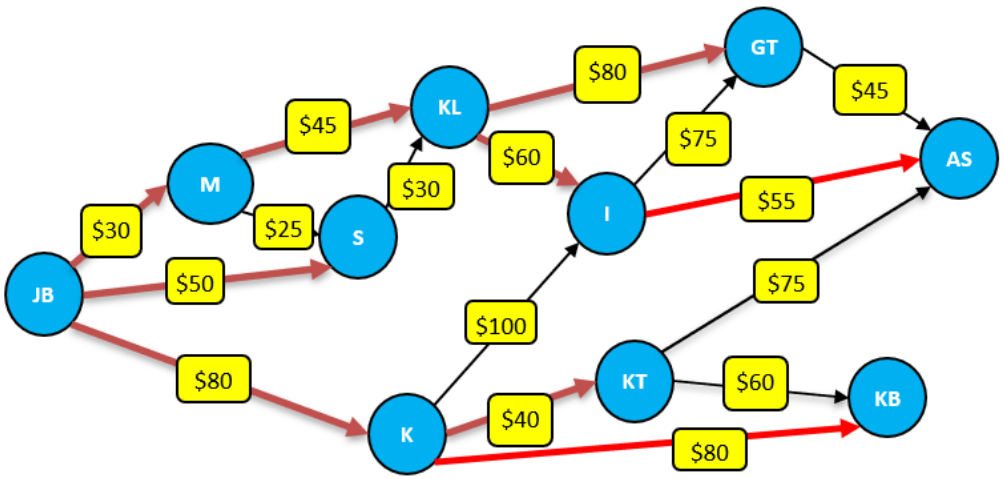
Iteration	Solution Set	Arcs	Total Cost
Iteration 3	JB M (30) S (50)	JB -> K **M -> KL** S -> KL	80 **75** 50+30 = 80

Iteration	Solution Set	Arcs	Total Cost
Iteration 4	JB M (30) S (50) KL (75)	**JB - K** KL - I KL - GT	**80** 75+60 = 135 75+80 = 155

Iteration	Solution Set	Arcs	Total Cost
Iteration 5	JB M (30) S (50) KL (75) K (80)	KL - I KL - GT **K - KT** K - I K - KB	75+60 = 135 75+80 = 155 **80+40 = 120** 80+100 = 180 80+80 = 160

Iteration	Solution Set	Arcs	Total Cost
Iteration 6	JB M (30) S (50) KL (75) K (80) KT (120)	**KL - I** KL - GT K - I K - KB KT - KB KT - AS	**75+60 = 135** 75+80 = 155 80+100 = 180 80+80 = 160 120+60 = 180 120+75 = 195

Dijkstra's Algorithm



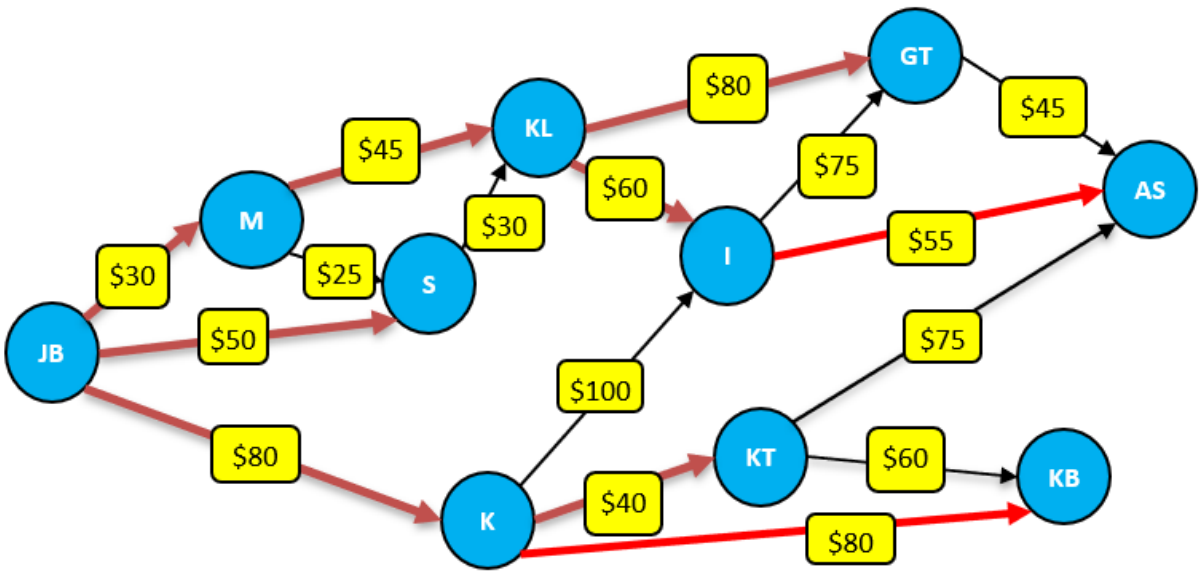
Iteration	Solution Set	Arcs	Total Cost
Iteration 7	JB M (30) S (50) KL (75) K (80) KT (120) I (135)	**KL - GT** K - KB KT - KB KT - AS I - AS I - GT	**75+80 = 155** 80+80 = 160 120+60 = 180 120+75 = 195 135+65 = 200 135+75 = 210

Iteration	Solution Set	Arcs	Total Cost
Iteration 8	JB M (30) S (50) KL (75) K (80) KT (120) I (135) GT (155)	**K - KB** KT - KB KT - AS I - AS GT - AS	**80+80 = 160** 120+60 = 180 120+75 = 195 135+55 = 190 155+45 = 200

Iteration	Solution Set	Arcs	Total Cost
Iteration 9	JB M (30) S (50) KL (75) K (80) KT (120) I (135) GT (155) KB (160)	KT - AS **I - AS** GT - AS	120+75 = 195 **135+55 = 190** 155+45 = 200

Iteration	Solution Set	Arcs	Total Cost
Iteration 10	JB M (30) S (50) KL (75) K (80) KT (120) I (135) GT (155) KB (160) AS (190)		

The Shortest Path Solution

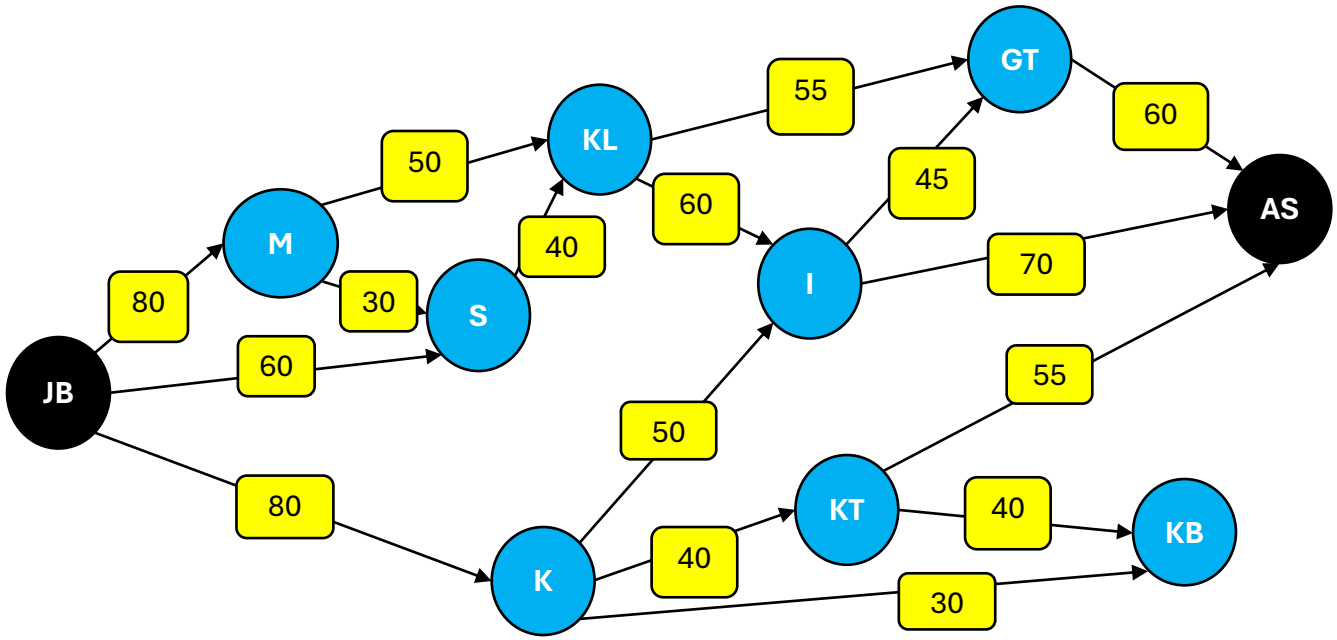


From JB to	Shortest Path	Total Cost
M	JB -> M	30
S	JB -> S	50
KL	JB -> M -> KL	75
K	JB -> K	80
KT	JB -> K -> KT	120
I	JB -> M -> KL -> I	135
GT	JB -> M -> KL -> GT	155
KB	JB -> K -> KB	160
AS	JB -> M -> KL -> I -> AS	190

<u>Assumptions</u>

- Unit cost of shipment does not change with shipment quantity.
- Routes in the network are fixed.
- There is no limitation of route capacity.

The Maximum Flow Problem



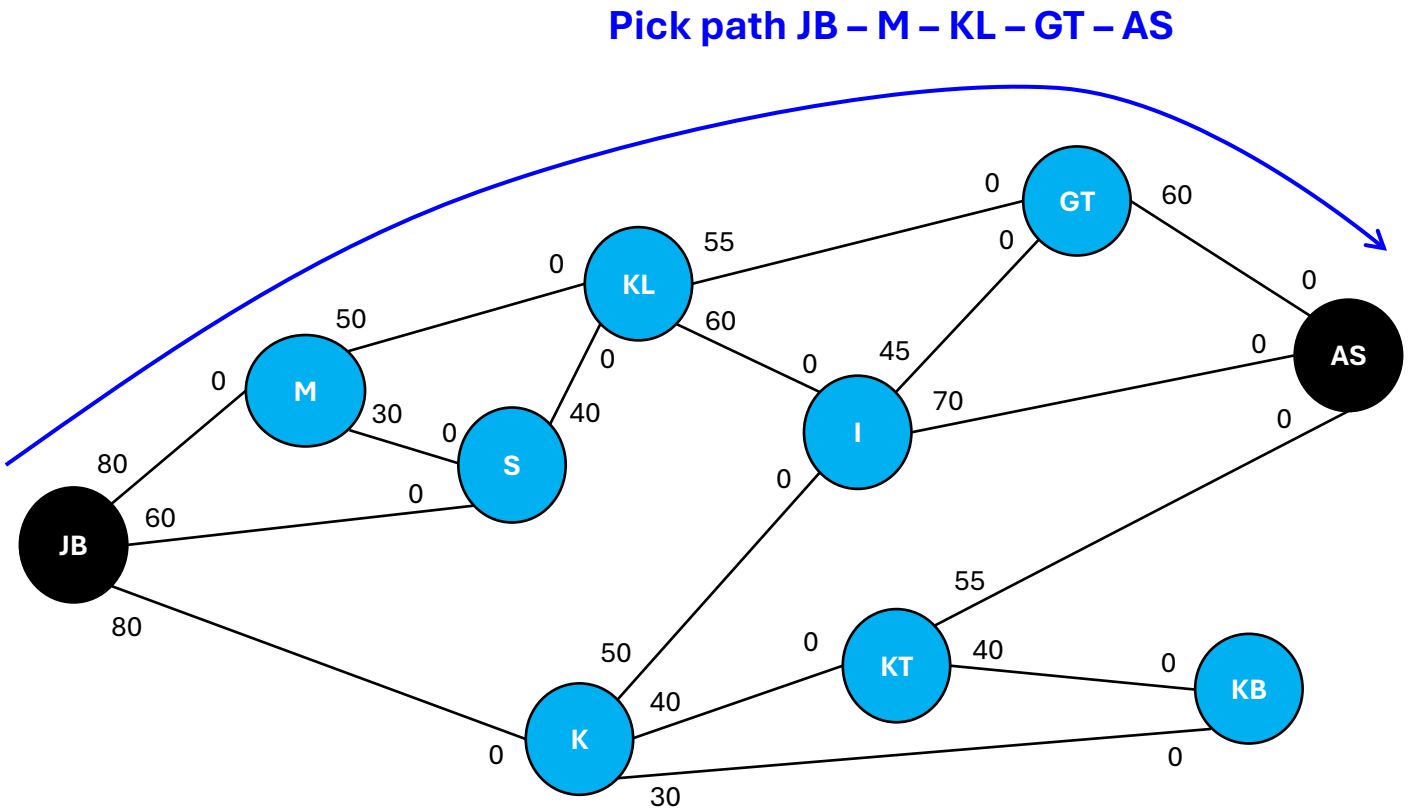
- To determine the maximum flow of products from Johor Bahru (Origin) to Alor Setar (Destination), and the amount to deliver along each route, we apply Ford-Fulkerson's Algorithm.

- Assumptions:
  - Cost of delivery is not a factor in determining the routes
  - Products are only delivered from Johor to Alor Setar
  - Delivery routes and capacity are fixed

Ford Fulkerson's Algorithm

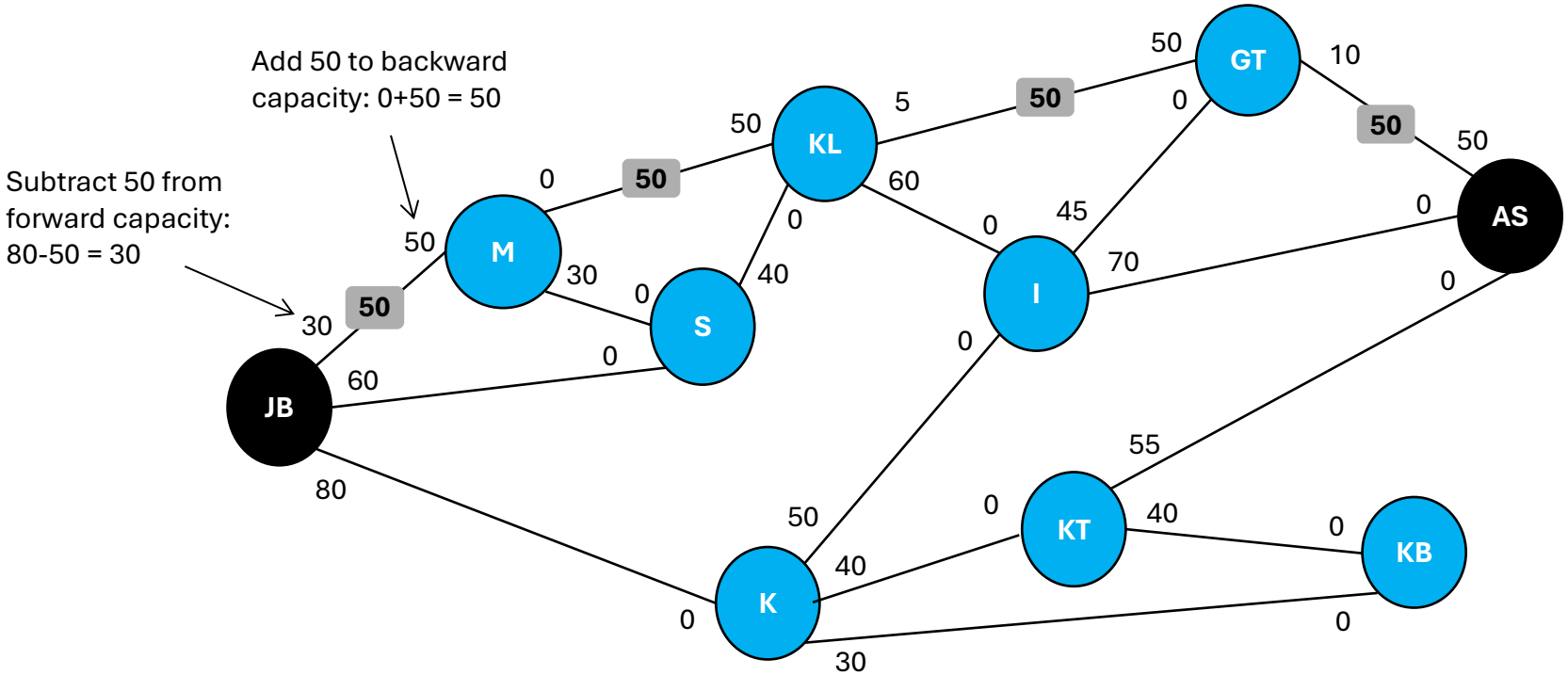
Solution Step 1

Convert routes to non-directional arcs and write down forward and reverse arc capacities.



Ford Fulkerson's Algorithm

Solution Step 2: Work out Flow Quantity and Balance Capacities



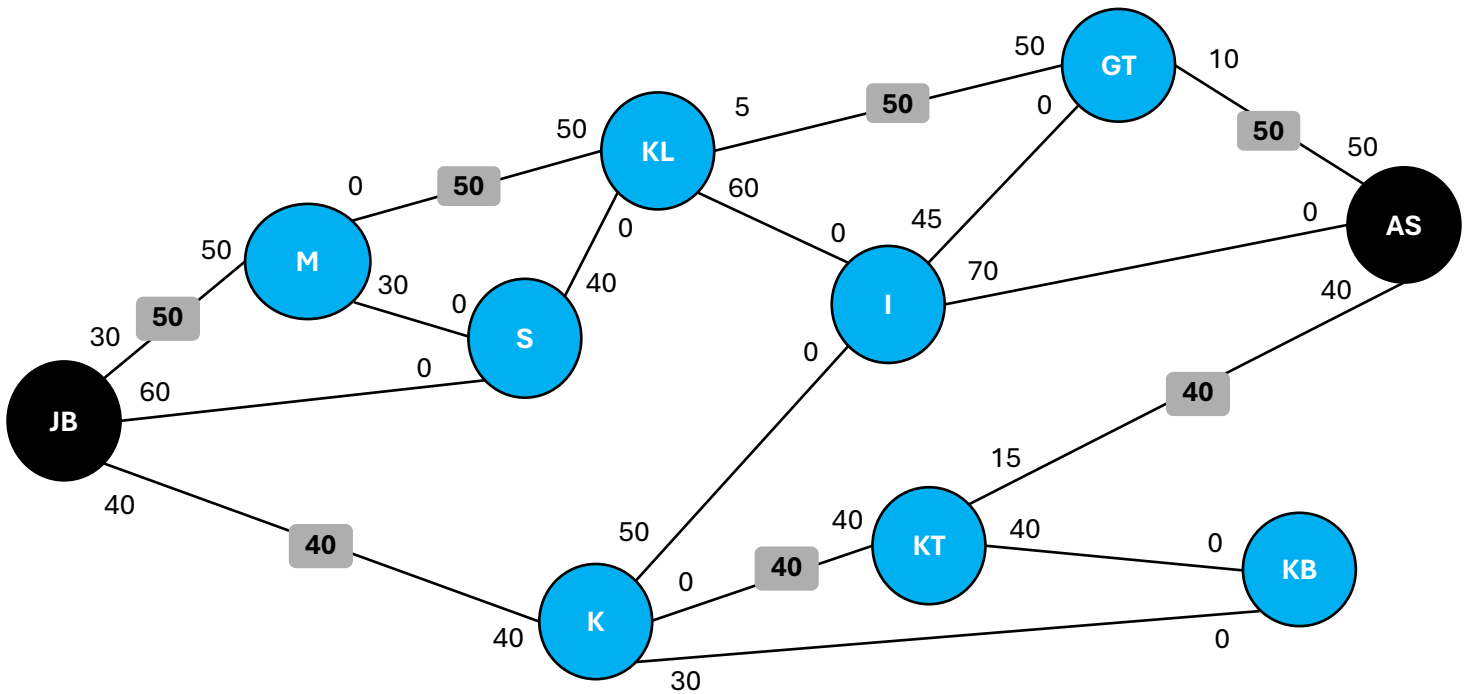
Route	Flow Quantity
JB - M - KL - GT - AS	50

Flow quantity = **smallest** capacity of all arcs along the route

Next, pick path JB – K – KT – AS

Ford Fulkerson's Algorithm

Solution Step 3: Repeat workings for Path JB – K – KT – AS

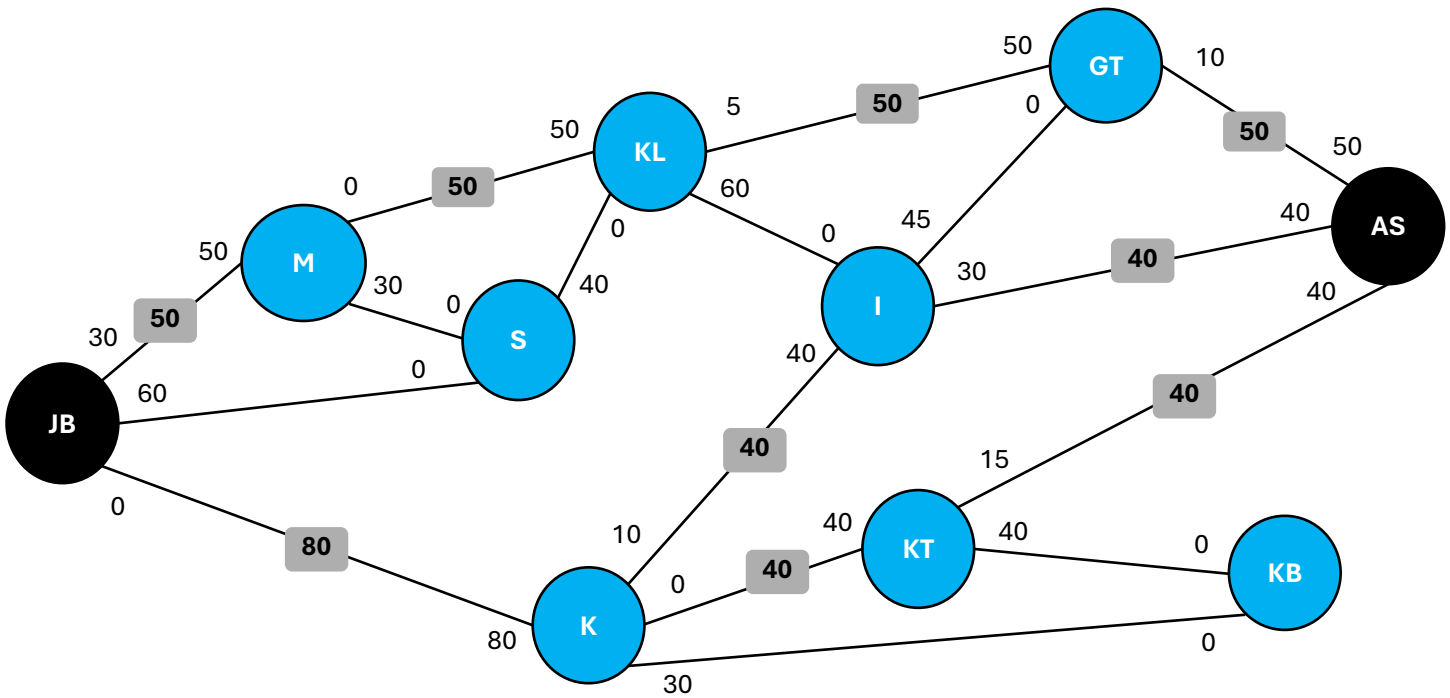


Route	Flow Quantity
JB - M - KL - GT - AS	50
JB - K - KT - AS	40

Next, pick path JB – K – I – AS

Ford Fulkerson's Algorithm

Solution Step 4: Repeat workings for Path JB – K – I – AS

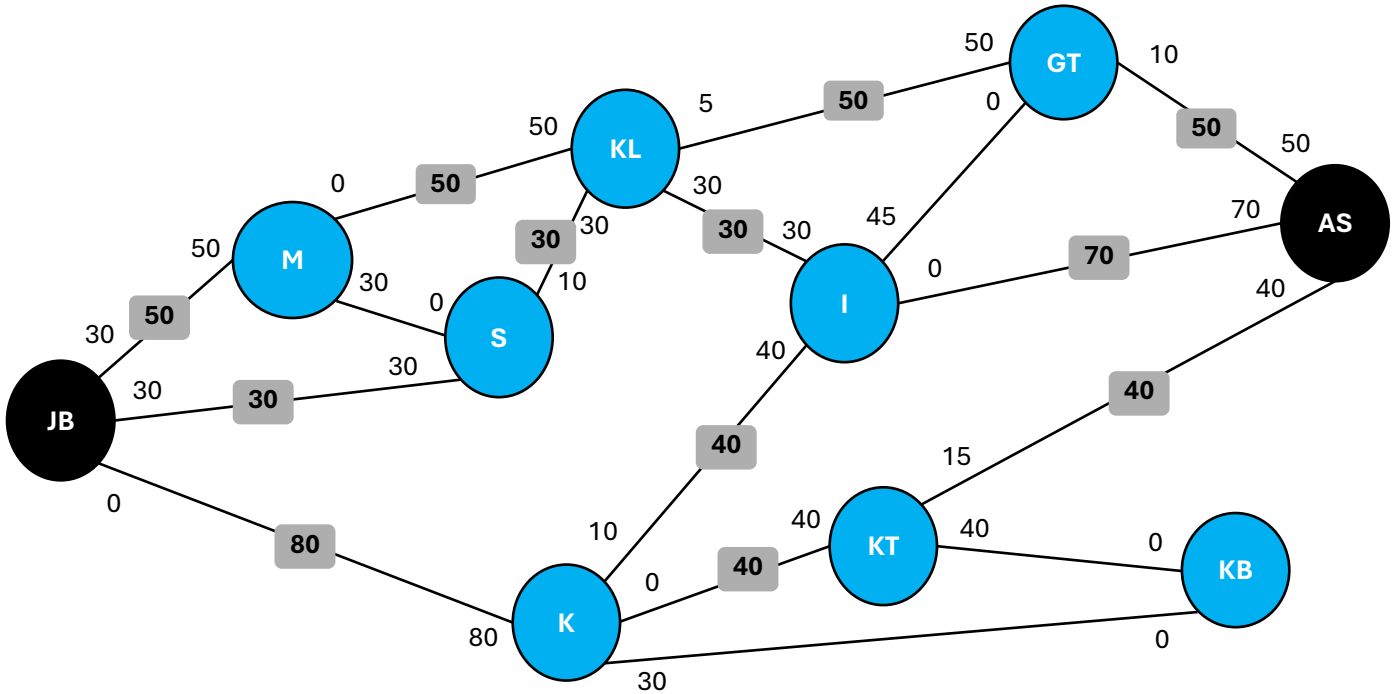


Route	Flow Quantity
JB - M - KL - GT - AS	50
JB - K - KT - AS	40
JB - K - I - AS	40

Next, pick path JB – S – KL – I – AS

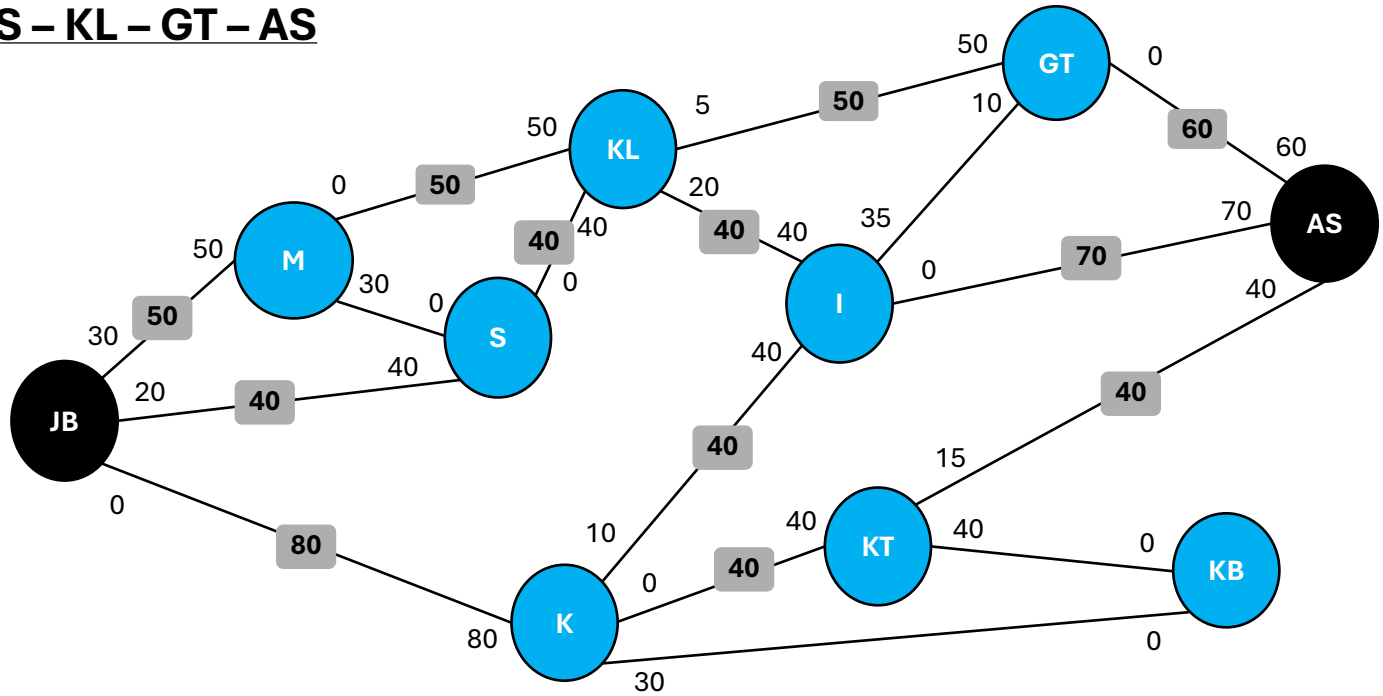
Ford Fulkerson's Algorithm

Solution Step 5: Repeat workings for Path JB – S – KL – I – AS


JB
M
S
KL
GT
I
AS
K
KT
KB
50
30
30
80
50
50
30
30
30
50
40
40
40
70
40
0
0
0

Ford Fulkerson's Algorithm

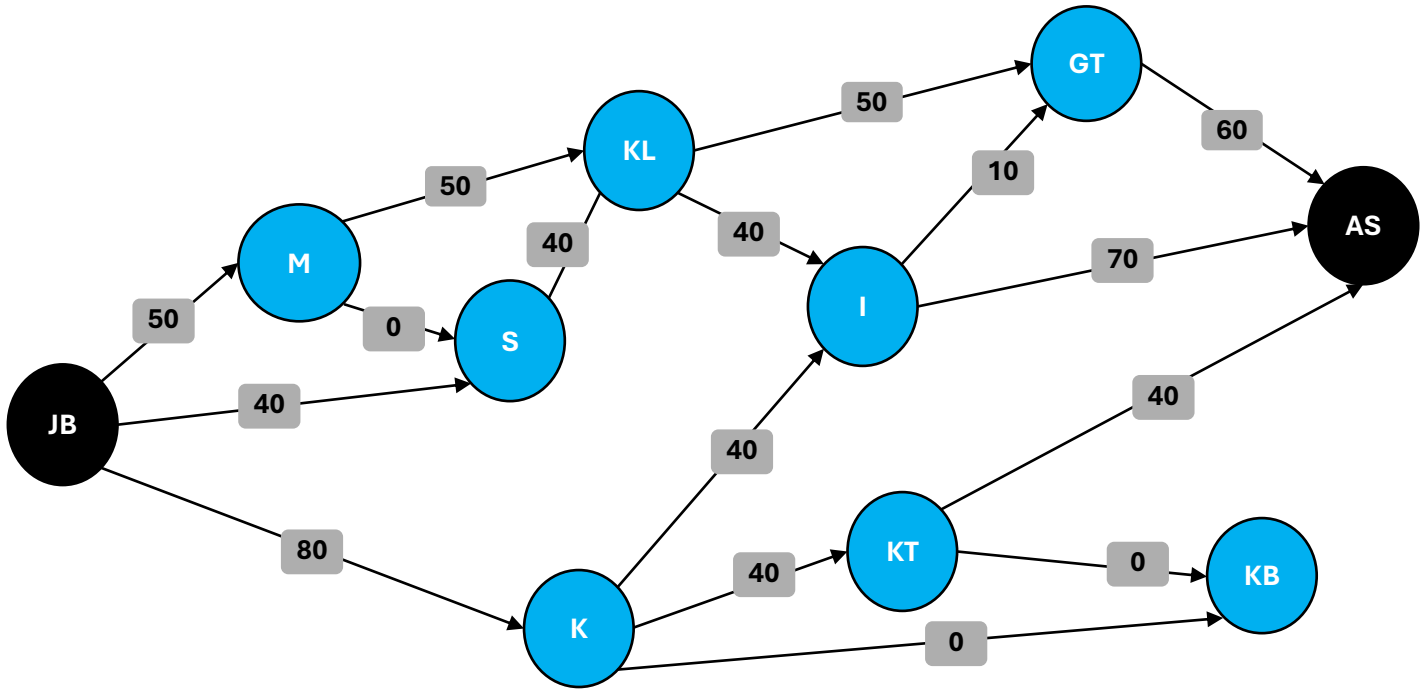
Solution Step 6: Repeat workings for Path JB – S – KL – GT – AS



Route	Flow Quantity
JB - M - KL - GT - AS	50
JB - K - KT - AS	40
JB - K - I - AS	40
JB - S - KL - I - AS	30
JB - S - KL - GT - AS	10

At this stage, there is no other path with capacity that is available.

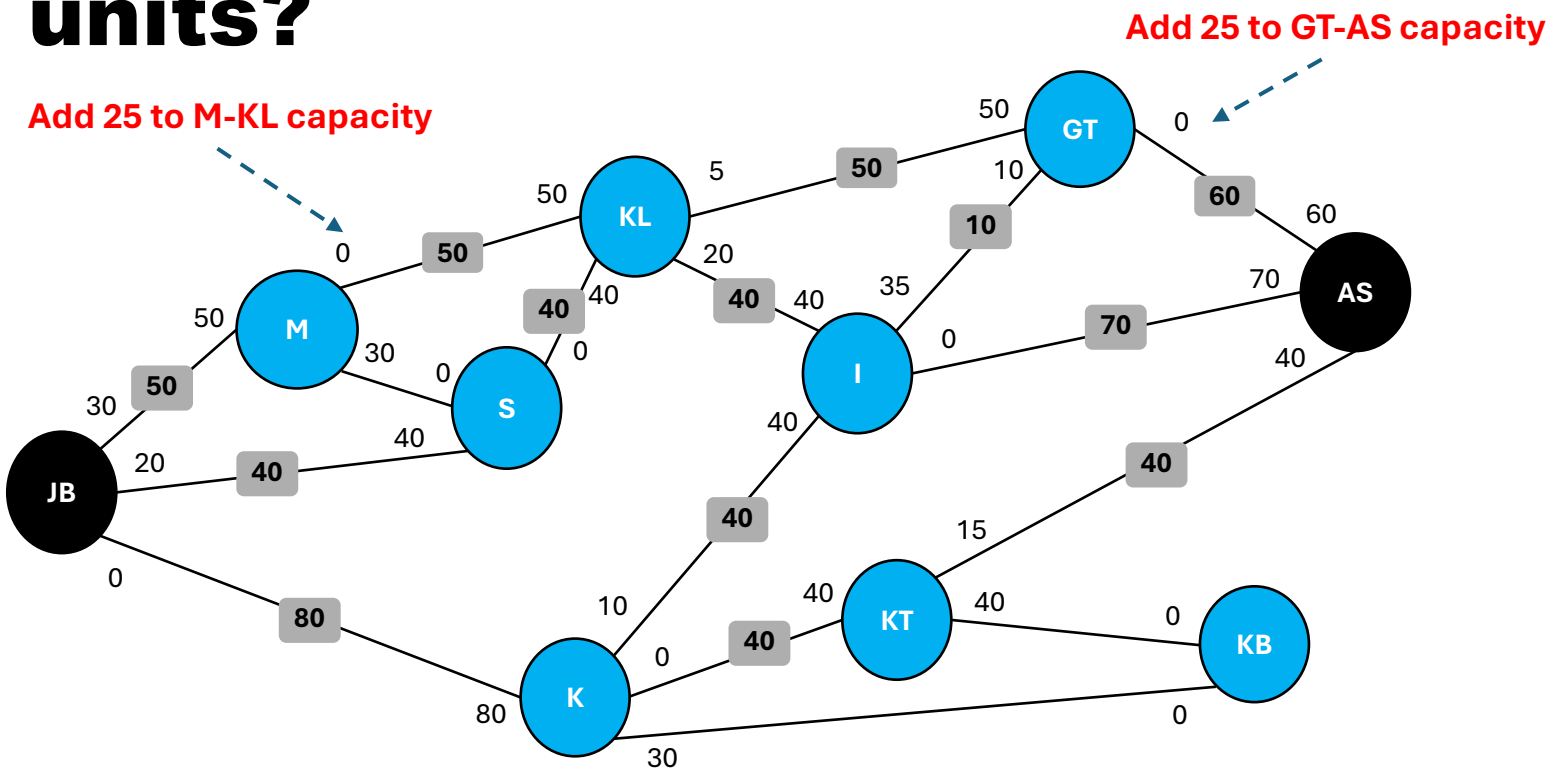
The Maximum Flow Solution



Route	Flow Quantity
JB - M - KL - GT - AS	50
JB - K - KT - AS	40
JB - K - I - AS	40
JB - S - KL - I - AS	30
JB - S - KL - GT - AS	10
Max Flow	**170**

There are multiple optimal solutions for this problem since the solution depends on selection of routes in the algorithm.

How to increase max flow by 25 units?



Increase max flow by 25 by increasing the capacity of:

- Arc M-KL by 25
- Arc GT-AS by 25

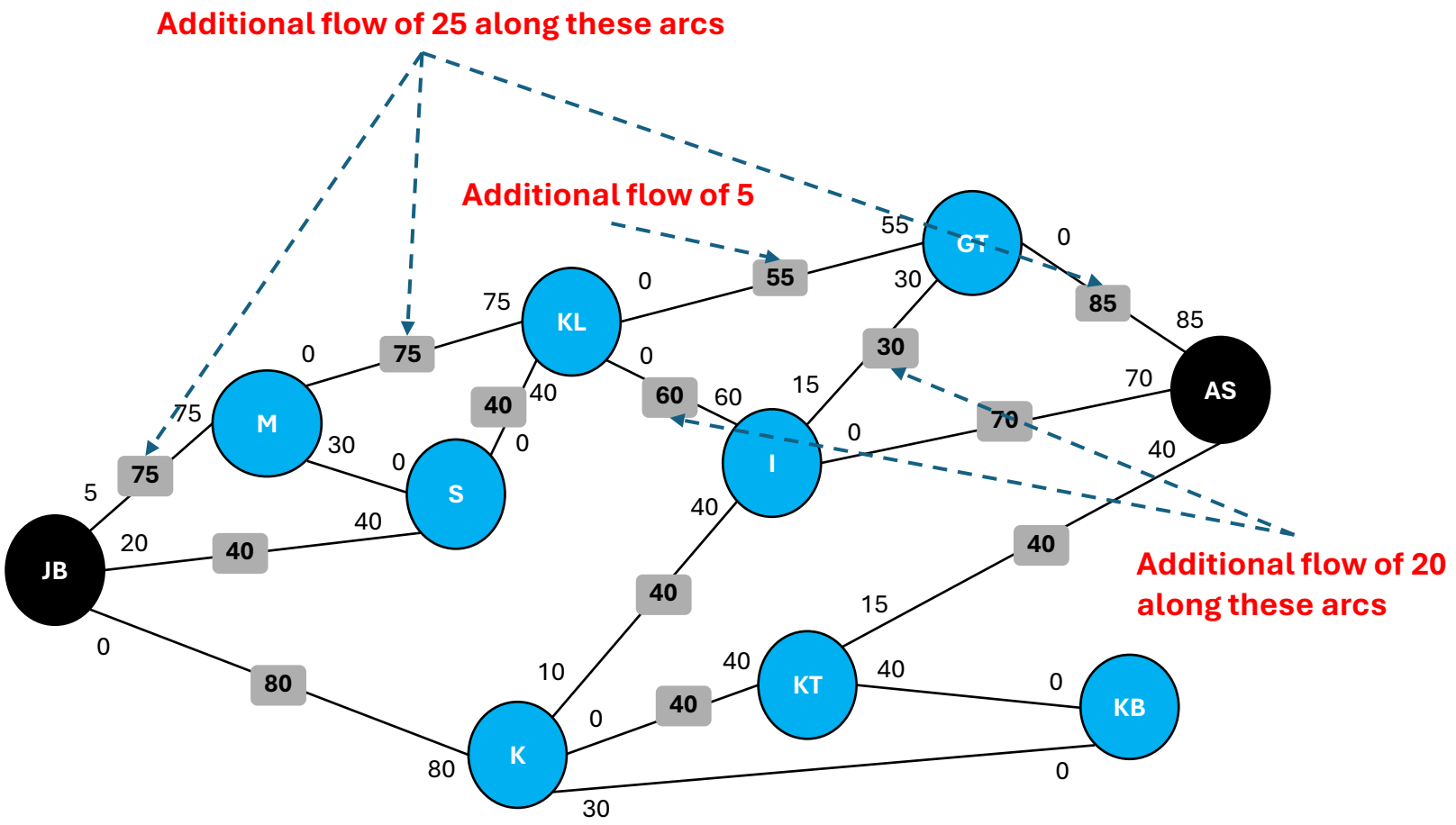
Additional network flow qty of 25 via:

JB - M - KL - GT - AS [5]

&

JB - M - KL - I - GT - AS [20]

Increase max flow by 25



Summary

- The cheapest delivery routes from Johor Bahru to each of the other 9 cities can be determined by solving it as a Shortest Path Problem.

- The maximum flow of products from Johor Bahru to Alor Setar and the amount of delivery along each route can be determined by solving it as a Maximum Flow Problem.

- Dijkstra's Algorithm and Ford-Fulkerson's Algorithm are used to solve Shortest Path and Maximum Flow Problem respectively. Both methods give optimal solutions.

Lesson 18 Solution

Nearest neighbour method and Clarke & Wright savings method

Learning Outcomes

- Identify what is a Vehicle Routing and Scheduling problem: Objectives, constraints, characteristics and classification of problems.

- Apply heuristic algorithm: Nearest Neighbour Method to obtain a feasible route.

- Apply heuristic algorithm: Clarke and Wright Savings method to obtain a feasible route.

The Nearest Neighbor Method Solution

From		To (Distance in km)								
		A	B	C	D	E	F	G	H	I
Serangoon North (Depot)	A	0	8.1	6	6	5.2	4.3	3.8	4	4
Jalan Angklong	B	8.1	0	13.5	14.4	14.3	9.1	6.7	9.5	9.3
Sumang link	C	6	13.5	0	3.7	3.6	9.1	9.7	6.2	5.6
Rivervale Crescent	D	6	14.4	3.7	0	1.7	6.8	7.1	4.4	4.2
Rivervale Street	E	5.2	14.3	3.6	1.7	0	6.5	7.1	4.2	4.5
Jalan Pelikat	F	4.3	9.1	9.1	6.8	6.5	0	2.3	7.7	8.2
Lorong Lew Lian	G	3.8	6.7	9.7	7.1	7.1	2.3	0	7.2	6.6
Fernvale Link	H	4	9.5	6.2	4.4	4.2	7.7	7.2	0	1.3
Sengkang West Way	I	4	9.3	5.6	4.2	4.5	8.2	6.6	1.3	0

From	To	Distance
A	G	3.8
G	F	2.3
F	E	6.5
E	D	1.7
D	C	3.7
C	I	5.6
I	H	1.3
H	B	9.5
B	A	8.1
	Total Dist.	42.5 km

Route:

A -> G -> F -> E -> D -> C -> I -> H -> B -> A

How will the solution change if the starting node is another node?

Clarke and Wright Savings Method

From		To (Distance in km)								
		A	B	C	D	E	F	G	H	I
Serangoon North (Depot)	A	0	8.1	6	6	5.2	4.3	3.8	4	4
Jalan Angklong	B	8.1	0	13.5	14.4	14.3	9.1	6.7	9.5	9.3
Sumang link	C	6	13.5	0	3.7	3.6	9.1	9.7	6.2	5.6
Rivervale Crescent	D	6	14.4	3.7	0	1.7	6.8	7.1	4.4	4.2
Rivervale Street	E	5.2	14.3	3.6	1.7	0	6.5	7.1	4.2	4.5
Jalan Pelikat	F	4.3	9.1	9.1	6.8	6.5	0	2.3	7.7	8.2
Lorong Lew Lian	G	3.8	6.7	9.7	7.1	7.1	2.3	0	7.2	6.6
Fernvale Link	H	4	9.5	6.2	4.4	4.2	7.7	7.2	0	1.3
Sengkang West Way	I	4	9.3	5.6	4.2	4.5	8.2	6.6	1.3	0

Clarke and Wright Savings Table

From		Savings								
		A	B	C	D	E	F	G	H	I
Serangoon North (Depot)	A	-	-	-	-	-	-	-	-	-
Jalan Angklong	B	-	-	0.6	-0.3	-1	3.3	5.2	2.6	2.8
Sumang link	C	-	0.6	-	8.3	7.6	1.2	0.1	3.8	4.4
Rivervale Crescent	D	-	-0.3	8.3	-	9.5	3.5	2.7	5.6	5.8
Rivervale Street	E	-	-1	7.6	9.5	-	3	1.9	5	4.7
Jalan Pelikat	F	-	3.3	1.2	3.5	3	-	5.8	0.6	0.1
Lorong Lew Lian	G	-	5.2	0.1	2.7	1.9	5.8	-	0.6	1.2
Fernvale Link	H	-	2.6	3.8	5.6	5	0.6	0.6	-	6.7
Sengkang West Way	I	-	2.8	4.4	5.8	4.7	0.1	1.2	6.7	-

Calculate the Savings for linking each pair of nodes (other than Depot (A) node)

Savings Calculation (Example): $S_{BC} = D_{A-B} + D_{A-C} - D_{B-C}$

$$S_{BC} = 8.1 + 6 - 13.5 = 0.6$$

Clarke and Wright Savings Method

Clarke and Wright Savings Ranking & Tour Construction

Rank	Saving	From	To	Tour	Remarks
1	9.5	D	E	[1] **A-D-E-A**	Initiate new tour
2	8.3	C	D	[1] **A-C-D-E-A**	Insert C between A and D
3	7.6	C	E		
4	6.7	H	I	[1] **A-C-D-E-A** [2] **A-H-I-A**	Initiate new tour
5	5.8	D	I		
6	5.8	F	G	[1] **A-C-D-E-A** [2] **A-H-I-A** [3] **A-F-G-A**	Initiate new tour
7	5.6	D	H		
8	5.2	B	G	[1] **A-C-D-E-A** [2] **A-H-I-A** [3] **A-F-G-B-A**	Insert B between G and A
9	5.0	E	H	[1] **A-C-D-E-H-I-A** [2] **A-F-G-B-A**	Join E and H and merge the subtours
10	4.7	E	I		
11	4.4	C	I		
12	3.8	C	H		
13	3.8	D	F		
14	3.3	B	F		
15	3.0	E	F		
16	2.8	B	I	[1] **A-C-D-E-H-I-B-G-F-A**	Join B and I and merge the subtours
17	2.7	D	G		
18	2.6	B	H		

What are the 3 conditions for nodes linking?

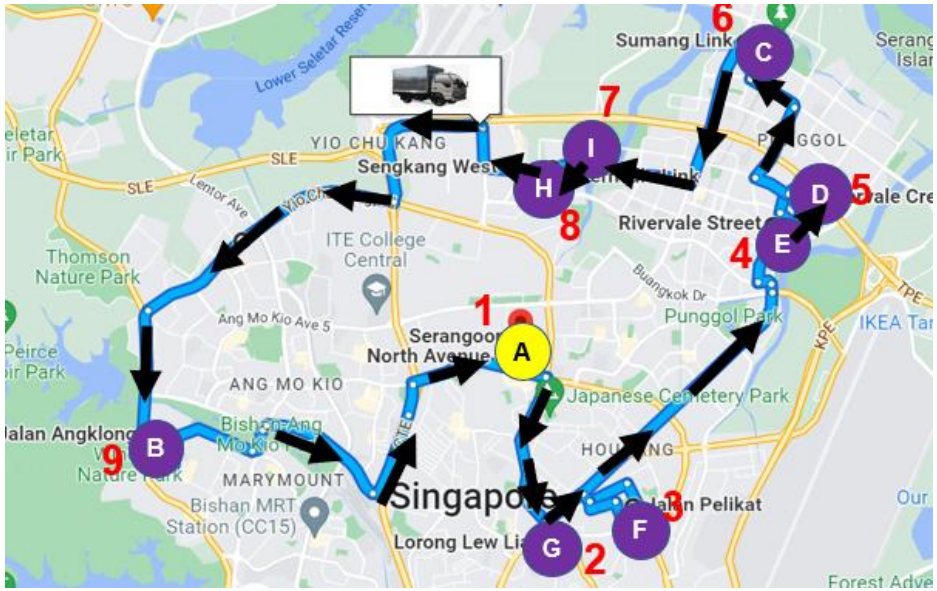
Route:

A -> C -> D -> E -> H -> I -> B -> G -> F -> A

From	To	Distance
A	C	6
C	D	3.7
D	E	1.7
E	H	4.2
H	I	1.3
I	B	9.3
B	G	6.7
G	F	2.3
F	A	4.3

Total Dist. 39.5 km

Nearest Neighbor Vs C&W



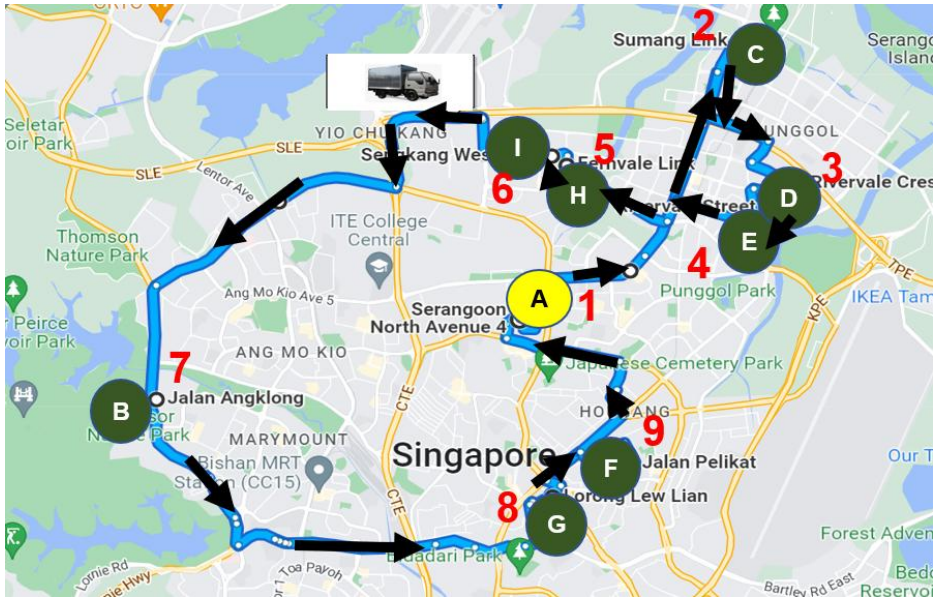
From	To	Distance
A	G	3.8
G	F	2.3
F	E	6.5
E	D	1.7
D	C	3.7
C	I	5.6
I	H	1.3
H	B	9.5
B	A	8.1

Total Dist. 42.5 km

Nearest Neigbourhood

Route:

A -> G -> F -> E -> D -> C -> I -> H -> B -> A



From	To	Distance
A	C	6
C	D	3.7
D	E	1.7
E	H	4.2
H	I	1.3
I	B	9.3
B	G	6.7
G	F	2.3
F	A	4.3

Total Dist. 39.5 km

Clarke & Wright Savings

Route:

A -> C -> D -> E -> H -> I -> B -> G -> F -> A

C&W Savings method provides shorter distance

With Vehicle Capacity Constraint

Clarke and Wright Savings Method

Rank	Saving	From	To	Tour	Remarks	Vehicle load	Comments
1	9.5	D	E	[1] **A-D-E-A**	Initiate new tour	[1] 150 + 200 = 350	
2	8.3	C	D	[1] **A-C-D-E-A**	Insert C between A and D	[1] 350 + 100 = 450	
3	7.6	C	E				
4	6.7	H	I	[1] **A-C-D-E-A** [2] **A-H-I-A**	Initiate new tour	[1] 450 [2] 180 + 90 = 270	
5	5.8	D	I				
6	5.8	F	G	[1] **A-C-D-E-A** [2] **A-H-I-A** [3] **A-F-G-A**	Initiate new tour	[1] 450 [2] 270 [3] 80 + 100 = 180	
7	5.6	D	H				
8	5.2	B	G	[1] **A-C-D-E-A** [2] **A-H-I-A** [3] **A-F-G-B-A**	Insert B between G and A	[1] 450 [2] 270 [3] 180 + 150 = 330	
9	5.0	E	H	[1] **A-C-D-E-A** [2] **A-H-I-A** [3] **A-F-G-B-A**		[1] 450 [2] 270 [3] 330	If E and H are joined, A-C-D-E-H-I-A load will be 720, which exceeds veh capacity
10	4.7	E	I				
11	4.4	C	I				
12	3.8	C	H				
13	3.8	D	F				
14	3.3	B	F				
15	3.0	E	F	[1] **A-C-D-E-A** [2] **A-H-I-A** [3] **A-F-G-B-A**		[1] 450 [2] 270 [3] 330	If E and F are joined, A-C-D-E-F-G-B-A load will be 800, which exceeds veh capacity
16	2.8	B	I	[1] **A-C-D-E-A** [2] **A-H-I-B-G-F-A**		[1] 450 [2] 270 + 330 = 600	Both tours' loads are within van capacity
17	2.7	D	G				

Customer	Delivery Load (kg)
B	150
C	100
D	150
E	200
F	80
G	100
H	180
I	90
Sum	1050

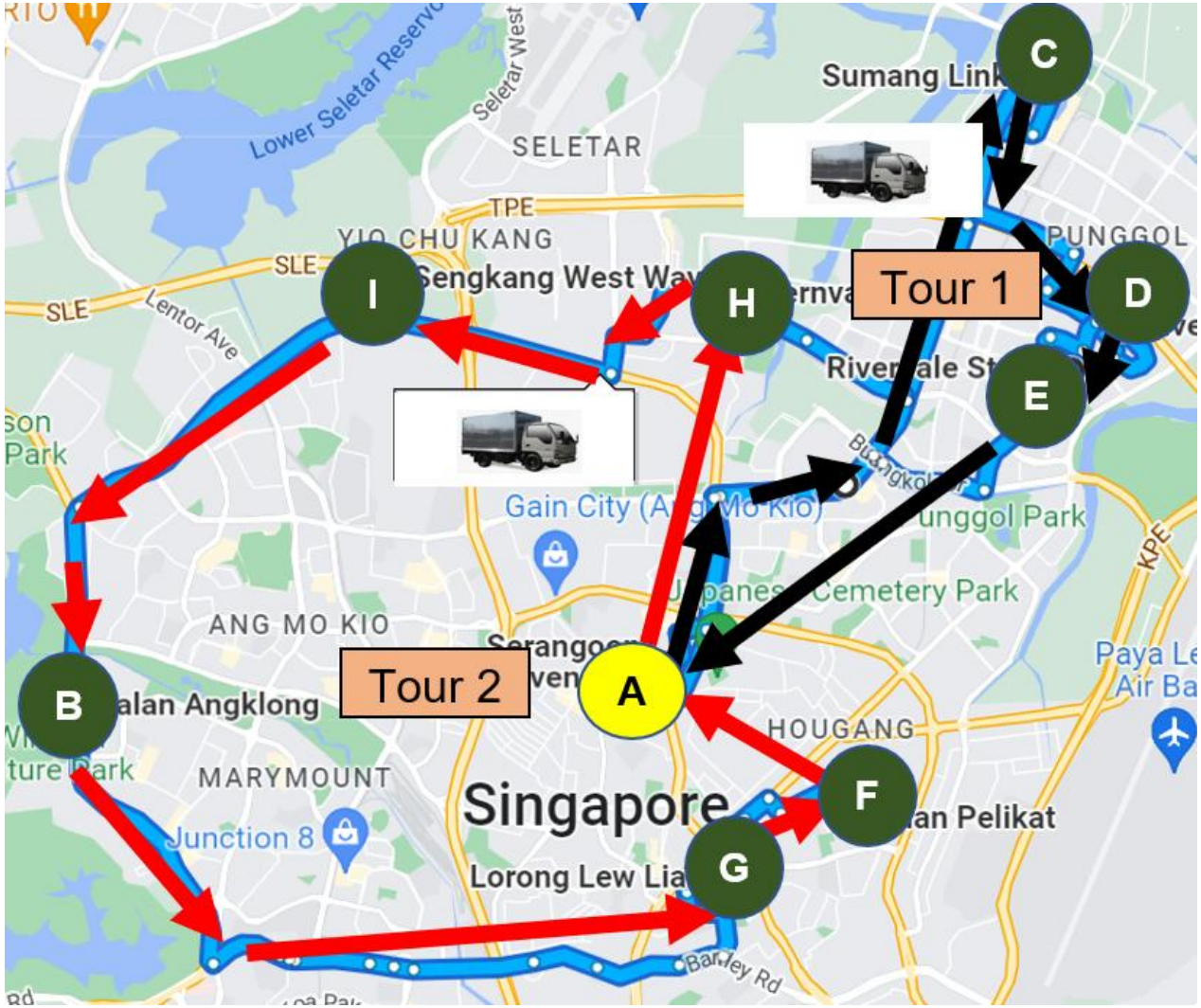
Tour [1]

From	To	Distance (km)	Delivery Load (kg)
A	C	6	100
C	D	3.7	150
D	E	1.7	200
E	A	5.2	
	Total	**16.6 km**	**450 kg**

Tour [2]

From	To	Distance (km)	Delivery Load (kg)
A	H	4	180
H	I	1.3	90
I	B	9.3	150
B	G	6.7	100
G	F	2.3	80
F	A	4.3	
	Total Dist.	**27.9 km**	**600 kg**

Route Visualization



Tour	Load (kg)	Distance (km)
[1] A-C-D-E-A	450	16.6
[2] A-H-I-B-G-F-A	600	27.9
Total	1050	44.5

Summary

- Clarke & Wright Savings Method will normally provide a better routing solution as compared with the Nearest Neighbour method. However, both methods do not guarantee optimal solutions.
- Clarke & Wright Savings Method is more tedious as compared to Nearest Neighbour method, which makes it more difficult to implement in a real application.
- If delivery time window is imposed at a certain location, the location's sequence would have to be scheduled to meet the time window constraint. For example, if delivery must be made at node F first, during tour construction, avoid inserting a node between Depot and node F.
- Delivery failure or unexpected delay in delivery can arise from a variety of reasons, such as customers not present during delivery, wrong address or wrong contact number so driver needs to do clarification, intercom is not working (e.g, for condos), language barrier in reading instructions for some drivers, traffic jam, bad weather, waiting for parking, rejection by customers, etc. There should be contingency plans for failed deliveries.

Lesson 20 Solution

Minimum cost flow problem

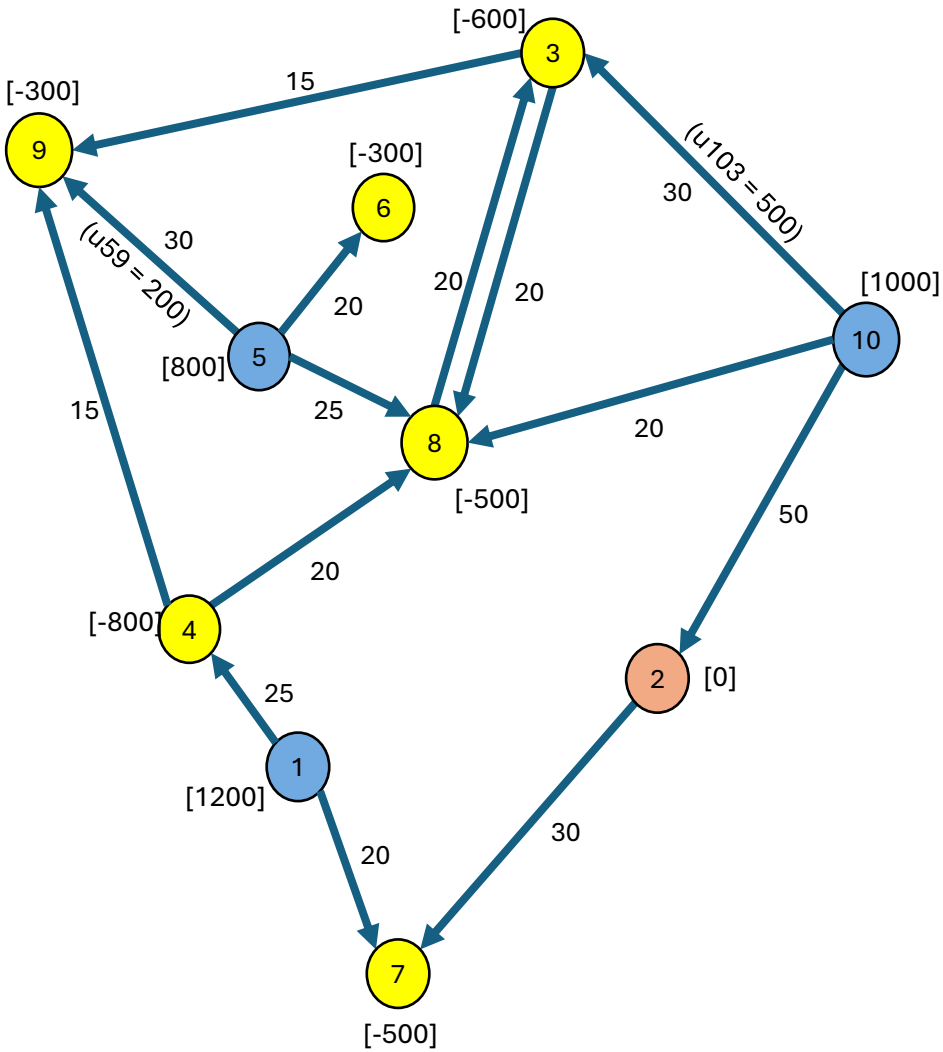
Learning Outcomes

- Explain that a distribution network problem deals with the distribution of goods from ‘sources’ to ‘destinations’ in a network with transhipment points and may consider cost of shipment and flow capacity.

- Formulate LP models for the minimum cost flow problem and shortest path problem.

- Solve LP models using Excel Solver for the minimum cost flow problem and shortest path problem.

Formulate a Minimum Cost Flow LP Model



Node	Location	Type of Node	Net flow
1	Singapore	Supply	1200
2	Bandar Seri Begawan	Transhipment	0
3	Hanoi	Demand	-600
4	Kuala Lumpur	Demand	-800
5	Bangkok	Supply	800
6	Vientiane	Demand	-300
7	Jakarta	Demand	-500
8	Phnom Penh	Demand	-500
9	Naypyidaw	Demand	-300
10	Manila	Supply	1000

Decision Variables & Objective Function

Decision Variables:

Let X_{ij} be the **number of units of toys t**o be shipped from node *i* to node *j*. In total, there are 14 **decision variables (same as the number of arcs)**:

X_{14}, $X_{17,}$ X_{27}, X_{38}, X_{39}, X_{48}, $X_{49,}$ X_{56}, X_{58}, X_{59}, X_{83}, X_{102}, X_{103}, X_{108}

Objective function:

Minimize total shipment cost, Z =

$25X_{14} + 20X_{17} + 30X_{27} + 20X_{38} + 15X_{39} + 20X_{48} + 15X_{49} + 20X_{56} + 25X_{58} + 30X_{59} + 20X_{83} + 50X_{102} + 30X_{103} + 20X_{108}$

Net Flow Constraints

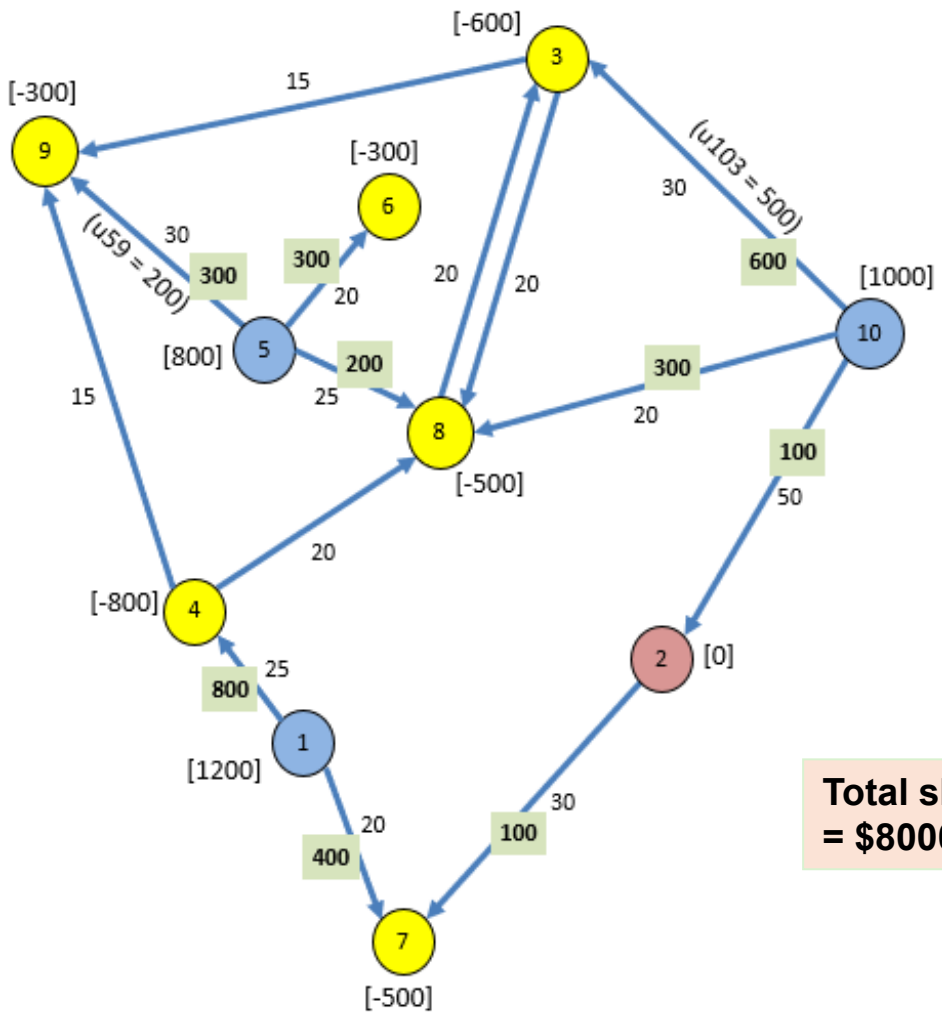
Subject to: Net Flow Constraints at Nodes

Supply Node 1:	x14 + x17	=	1000
Supply Node 5:	x56 + x58 + x59	=	900
Supply Node 10:	x102 + x103 + x108	=	1200
Transhipment Node 2:	x27 - x102	=	0
Demand Node 3:	x38 + x39 - x83 - x103	=	-700
Demand Node 4:	x48 + x49 - x14	=	-800
Demand Node 6:	- x56	=	-200
Demand Node 7:	- x17 - x27	=	-500
Demand Node 8:	x83 - x38 - x48 - x58 - x108	=	-500
Demand Node 9:	- x39 - x49 - x59	=	-400

Non-negativity constraint:

All X_{ij} >= 0

Solver Solution (No Arc Capacity Constraint)



Total shipment cost = $80000

Arc	Units
x14	800
x17	400
x27	100
x38	0
x39	0
x48	0
x49	0
x56	300
x58	200
x59	300
x83	0
x102	100
x103	600
x108	300

Arc Capacity Constraints

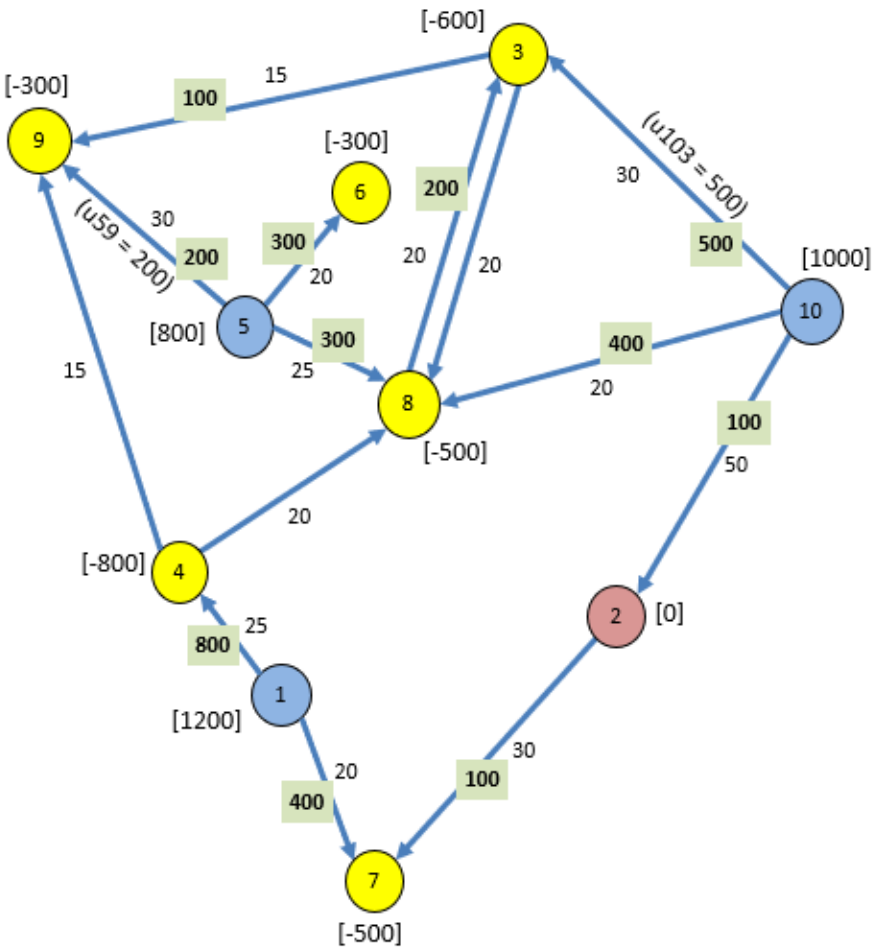
Shipping capacity is restricted to a maximum of 200 & 500 toy units from Bangkok (5) to Naypyidaw (9) and Manila (10) to Hanoi (3) respectively.

Arc capacity or Upper-Bound constraints

$X_{59} <= 200$

$X_{103} <= 500$

Solver Solution (with Arc Capacity Constraints)



Arc	Units
x14	800
x17	400
x27	100
x38	0
x39	100
x48	0
x49	0
x56	300
x58	300
x59	200
x83	200
x102	100
x103	500
x108	400

Total shipment cost = $84000

Total shipment cost is higher compared with no capacity constraint. The shipment capacity constraints result in the distribution becoming less optimal.

Total Supply ≠ Total Demand

How would the constraints in the LP model change if:

a) The plant at Singapore can only supply 1000 units of toys instead of 1200
b) The demand at Phnom Penh is reduced to 400 from 500

a) **Total supply < total demand**
→ All supplies are sent out whereas some demands are not fulfilled.

Change all demand node constraint signs to '>=' (because RHS values are negative).

Demand Node 3:	x38 + x39 - x83 - x103	>=	-600
Demand Node 4:	x48 + x49 - x14	>=	-800
Demand Node 6:	- x56	>=	-300
Demand Node 7:	- x17 - x27	>=	-500
Demand Node 8:	x83 - x38 - x48 - x58 - x108	>=	-500
Demand Node 9:	- x39 - x49 - x59	>=	-300
Supply Node 1:	x14 + x17	=	1000

b) **Total supply > total demand**
→ Some supplies are not sent out whereas all demands are met.

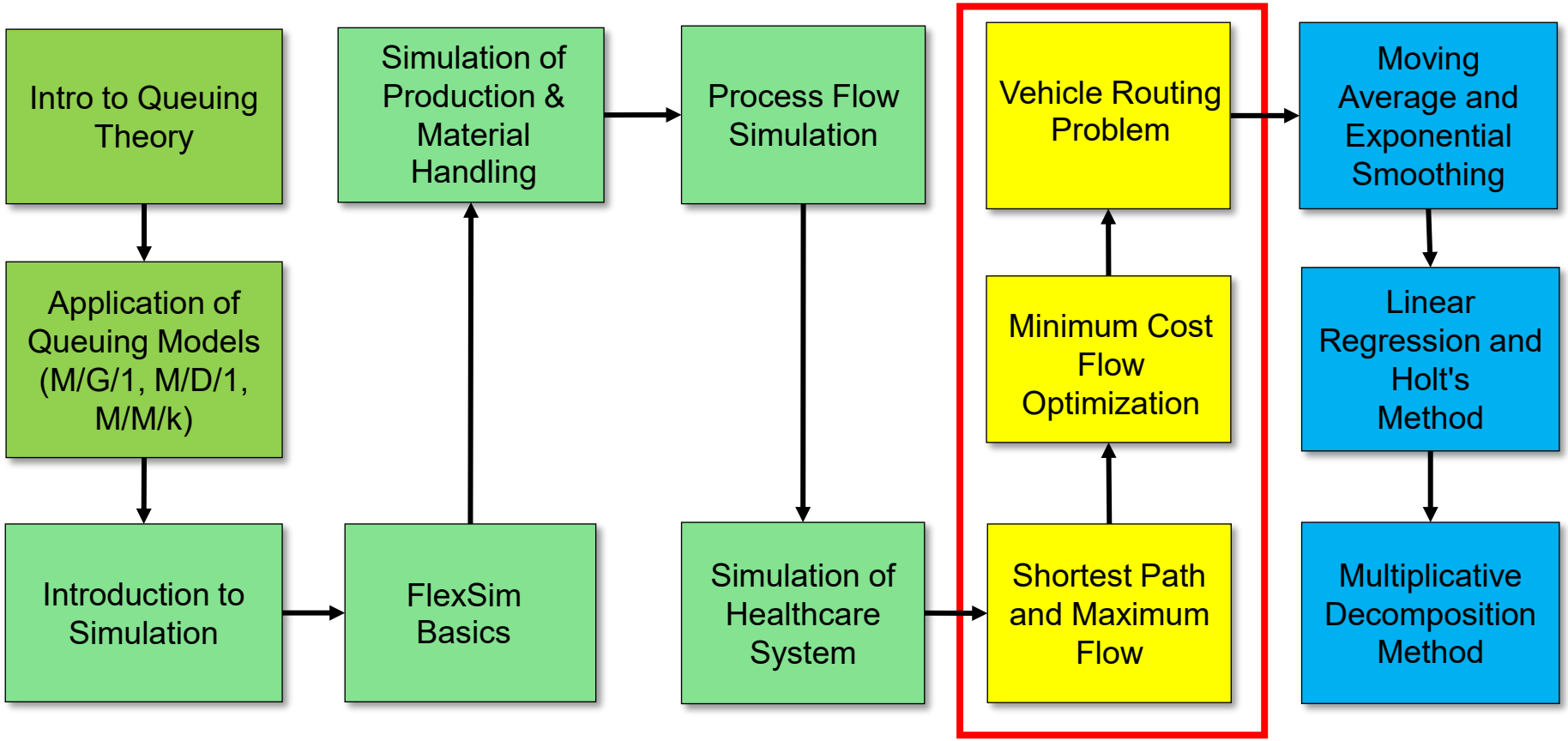
Change all supply node constraint signs should be to '<='

Supply Node 1:	x14 + x17	<=	1200
Supply Node 5:	x56 + x58 + x59	<=	800
Supply Node 10:	x102 + x103 + x108	<=	1000
Demand Node 8:	x83 - x38 - x48 - x58 - x108	=	-400

Summary

- The minimum cost flow problem holds a central position in the network optimization models.
- The transportation problem, shortest path problem, and maximum flow problem are all special cases of the minimum cost flow problem.
- One special application of the minimum cost flow problem is the distribution network problem.
- The distribution network problem is to find the minimum cost shipment arrangements distributing products from ‘sources’ to ‘destinations’ considering both transshipment points and flow capacity in a transportation network.
- Flow networks can also be used to model liquids flowing through pipes, parts through assembly lines, current through electrical networks, information through communication networks.

E211 Progression



L26 WS Solution

Demand Forecasting

E211 – Operations Planning II

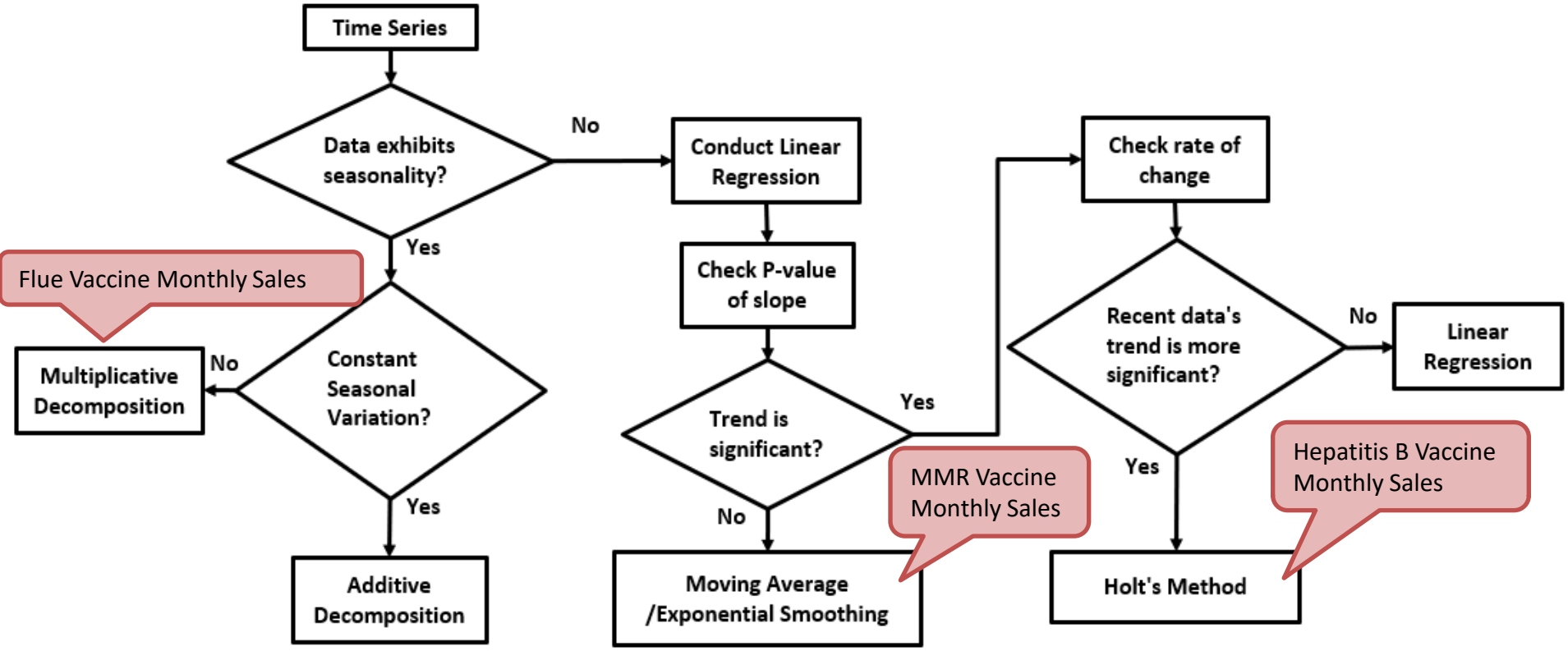
SCHOOL OF ENGINEERING

Selection of Forecasting Methods

1. When the data exhibits correlation between two variables,
   - Forecast of a dependant variable using one independent variable can be done through Linear Regression.

For example, use MMR Vaccine Monthly Sales (independent variable) to forecast MMR Vaccine Monthly Distribution Expenses (dependent variable).

2. When the data is a time series, use the following flowchart as a guide to choose the forecasting methods.



Forecast of MMR Vaccine Monthly Distribution Expenses based on its Monthly Sales

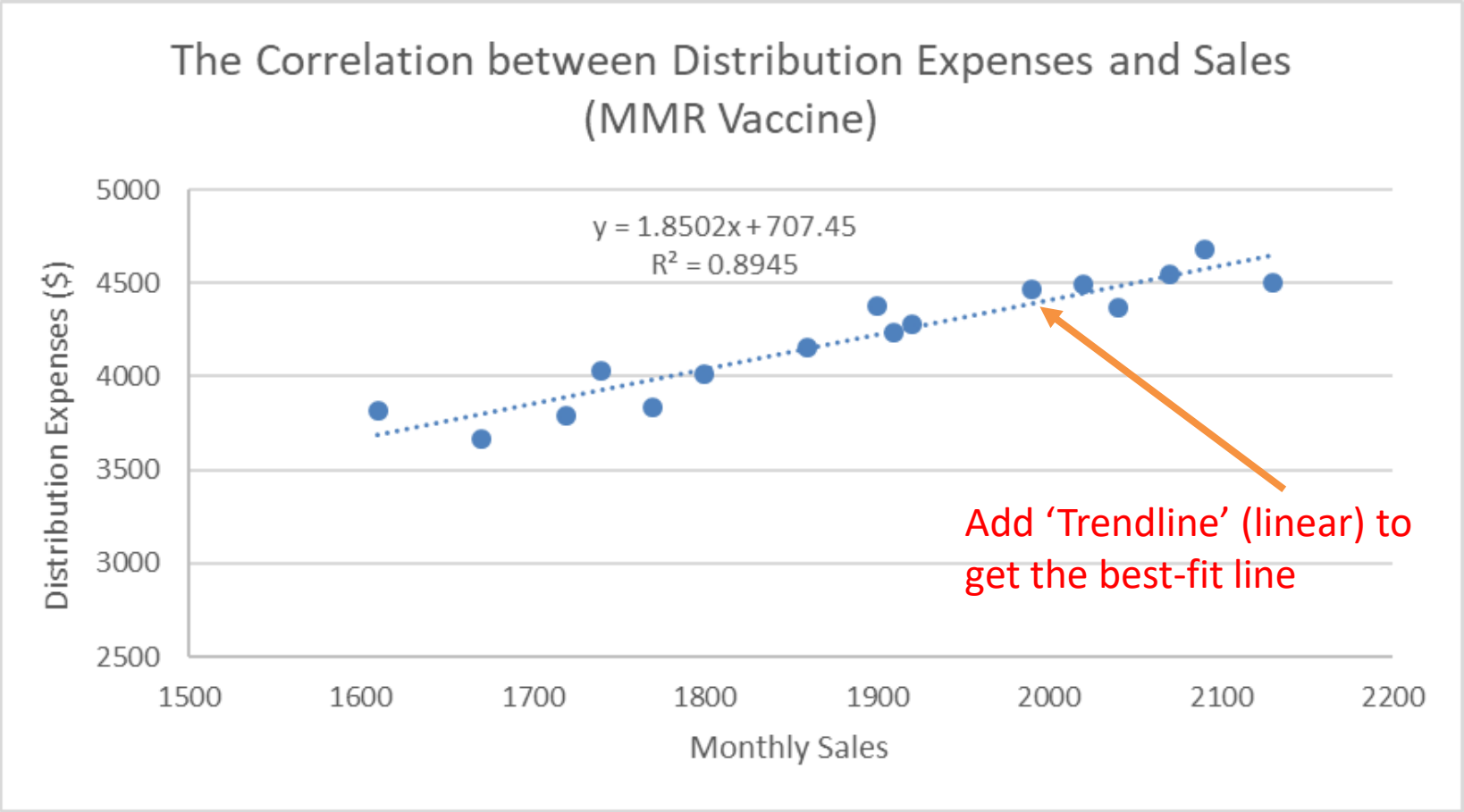
Causal Forecasting: Using MMR Vaccine Monthly Sales to Forecast MMR Vaccine Monthly Distribution Expenses

- Linear regression model:

$$y = \beta_0 + \beta_1 x$$

  - y – MMR Vaccine Monthly Distribution Expenses;
  - x – MMR Vaccine Monthly Sales (Sales Figure);
  - β_0 - intercept value;
  - β_1 - slope; the change in MMR Vaccine Monthly Distribution Expenses due to 1 unit change in MMR Vaccine Monthly Sales: we expect a positive correlation ($\beta_1 > 0$).

Scatter Plot for MMR Vaccine Monthly Sales and MMR Vaccine Monthly Distribution Expenses



β_0 = 707.45
β_1 = 1.8502
R^2= 0.8945

Interpretation of *r* and R^2 values

Correlation coefficient, r = 0.9458

- With r close to 1, it indicates that there is a strong relationship between the MMR Vaccine Monthly Distribution Expenses and the MMR Vaccine Monthly Sales.
- As β_1 (slope) is positive, r is also positive. It indicates a positive relationship between the MMR Vaccine Monthly Distribution Expenses and the MMR Vaccine Monthly Sales: the higher the monthly sales, the higher the monthly distribution expenses.

Coefficient of determination, R^2 ('R Square') = r*r =0.8945

- 89.45% of the variation in the MMR Vaccine Monthly Distribution Expenses is explained by the variation in the MMR Vaccine Monthly Sales.
- The rest of the variation may be due to other factors such as labour cost, fuel price, etc.

Using Regression Model to Predict y

Regression model: $y = 1.8502x + 707.4535$

- If MMR Vaccine Monthly Sales = 1000,

  Predicted MMR Vaccine Monthly Distribution Expenses:

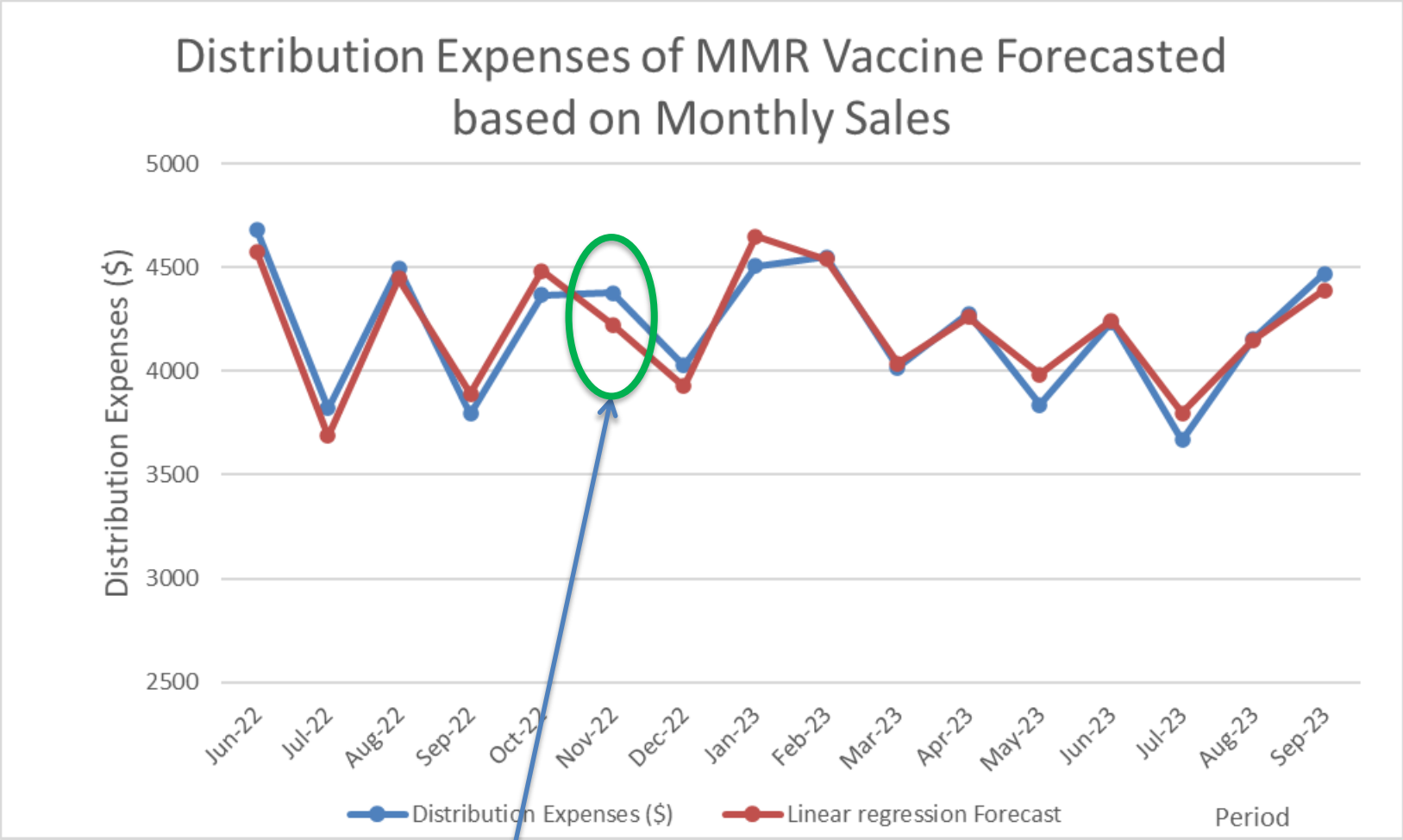
  $y = 1.8502*(1000) + 707.4535 = \$2557.65$

- If MMR Vaccine Monthly Sales = 2500,

  Predicted MMR Vaccine Monthly Distribution Expenses:

  $y = 1.8502*(2500) + 707.4535 = \$5332.95$

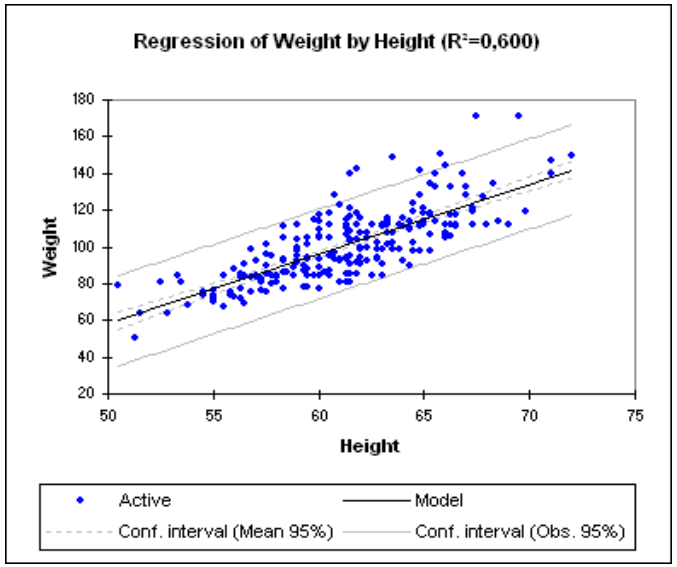
Using Linear Regression to Forecast MMR Vaccine Monthly Distribution Expenses based on MMR Vaccine Monthly Sales



- Other factors, such as labour cost, fuel price, etc., are affecting MMR Vaccine Monthly Distribution Expenses in certain periods in addition to MMR Vaccine Monthly Sales.

Using Linear Regression to Forecast MMR Vaccine Monthly Distribution Expenses based on MMR Vaccine Monthly Sales

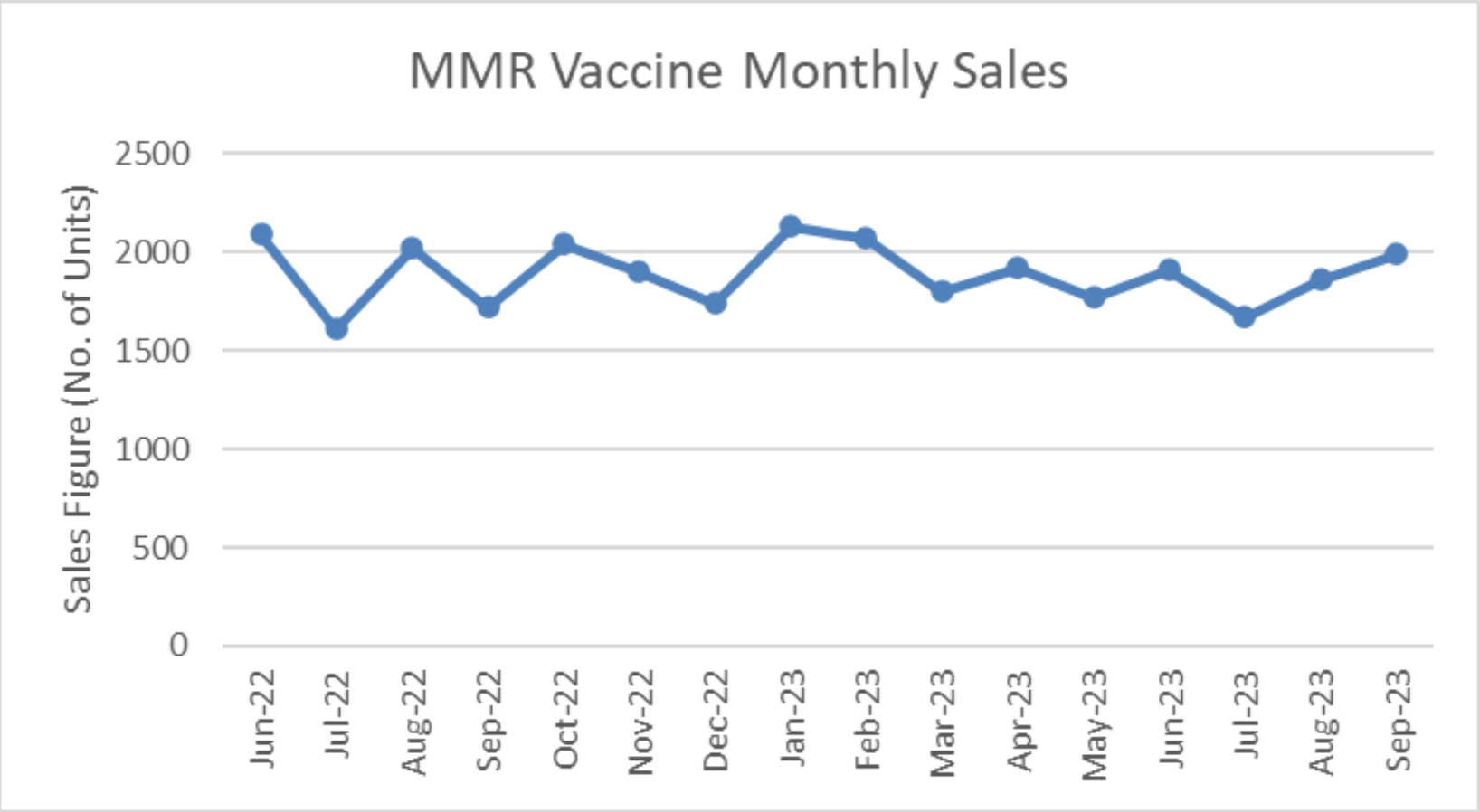
- If we already have a good estimate of MMR Vaccine Monthly Sales for the next month, linear regression may be a good method to forecast next month's MMR Vaccine Monthly Distribution Expenses as there is a strong correlation between MMR Vaccine Monthly Distribution Expenses and MMR Vaccine Monthly Sales.
- Ways to improve linear regression forecast:
  - Generate confidence intervals of predicted values to provide range estimates (upper and lower bounds).



  - Use more predictors (independent variables) – multiple regression model.

Forecast of MMR Vaccine Monthly Sales

MMR Vaccine Monthly Sales



The MMR Vaccine Monthly Sales data exhibits random up and down fluctuations.

MMR Vaccine Monthly Sales: Moving Average Solution

Period	Sales Figure (No. of Units)	MA, 3-Period	MA, 5-Period
1	2090		
2	1610		
3	2020		
4	1720	1906.67	
5	2040	1783.33	
6	1900	1926.67	1896.00
7	1740	1886.67	1858.00
8	2130	1893.33	1884.00
9	2070	1923.33	1906.00
10	1800	1980.00	1976.00
11	1920	2000.00	1928.00
12	1770	1930.00	1932.00
13	1910	1830.00	1938.00
14	1670	1866.67	1894.00
15	1860	1783.33	1814.00
16	1990	1813.33	1826.00

$$F_5 = \frac{1610+2020+1720}{3}$$

$$F_{12} = \frac{1740+2130+2070+1800+1920}{5}$$

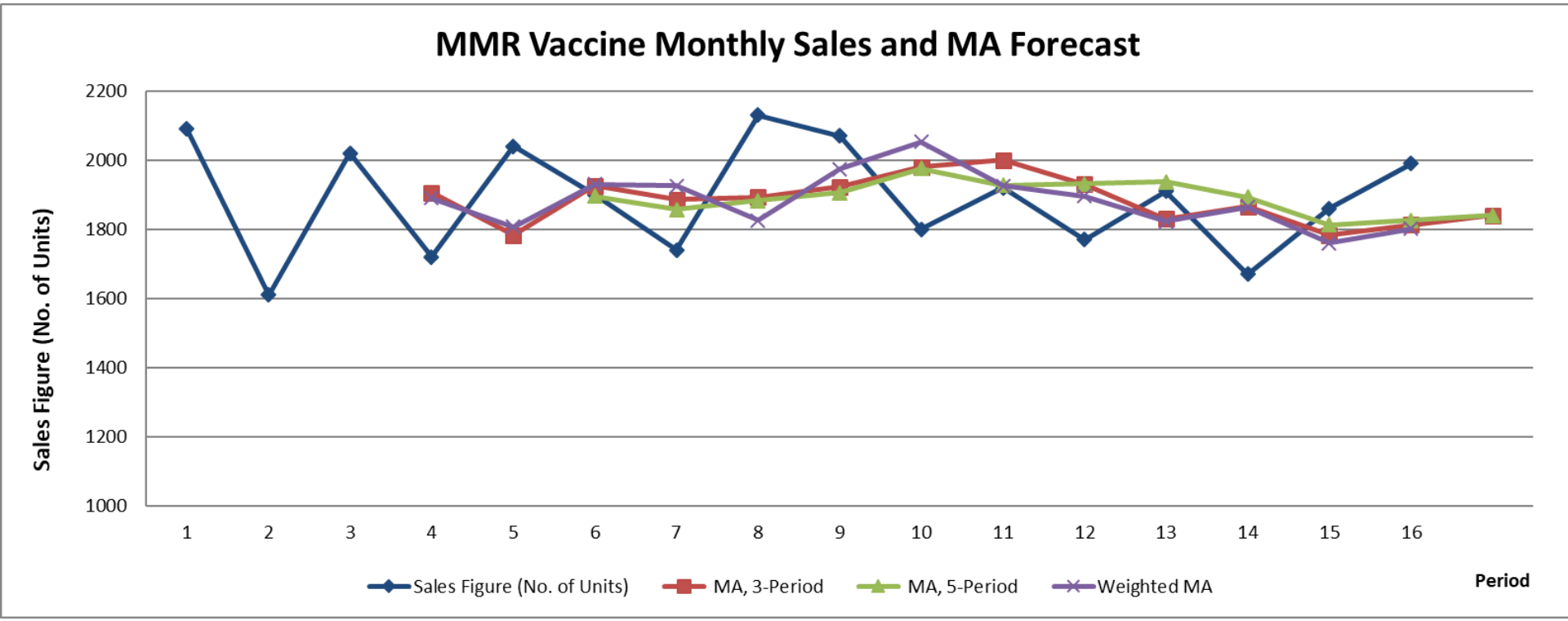
MMR Vaccine Monthly Sales: Weighted Moving Average

Period	Sales Figure (No. of Units)	**Weighted MA**
1	2090	
2	1610	
3	2020	
4	1720	1891.11
5	2040	1807.78
6	1900	1931.11
7	1740	1926.67
8	2130	1826.67
9	2070	1974.44
10	1800	2053.33
11	1920	1926.67
12	1770	1896.67
13	1910	1823.33
14	1670	1864.44
15	1860	1761.11
16	1990	1802.22

Weights assigned:
5 (most recent), 3, 1 (least recent)

$$F_7 = \frac{1900*5+2040*3+1720*1}{5+3+1}$$

MMR Vaccine Monthly Sales: Moving Average Solution



- Moving average forecast begins on the N+1 period (for 5 month MA, first forecast will be on 6th month).
- As N increases, the forecasted values become less responsive to changes in the actual data and graph becomes more smoothed.

MMR Vaccine Monthly Sales: Simple Exponential Smoothing

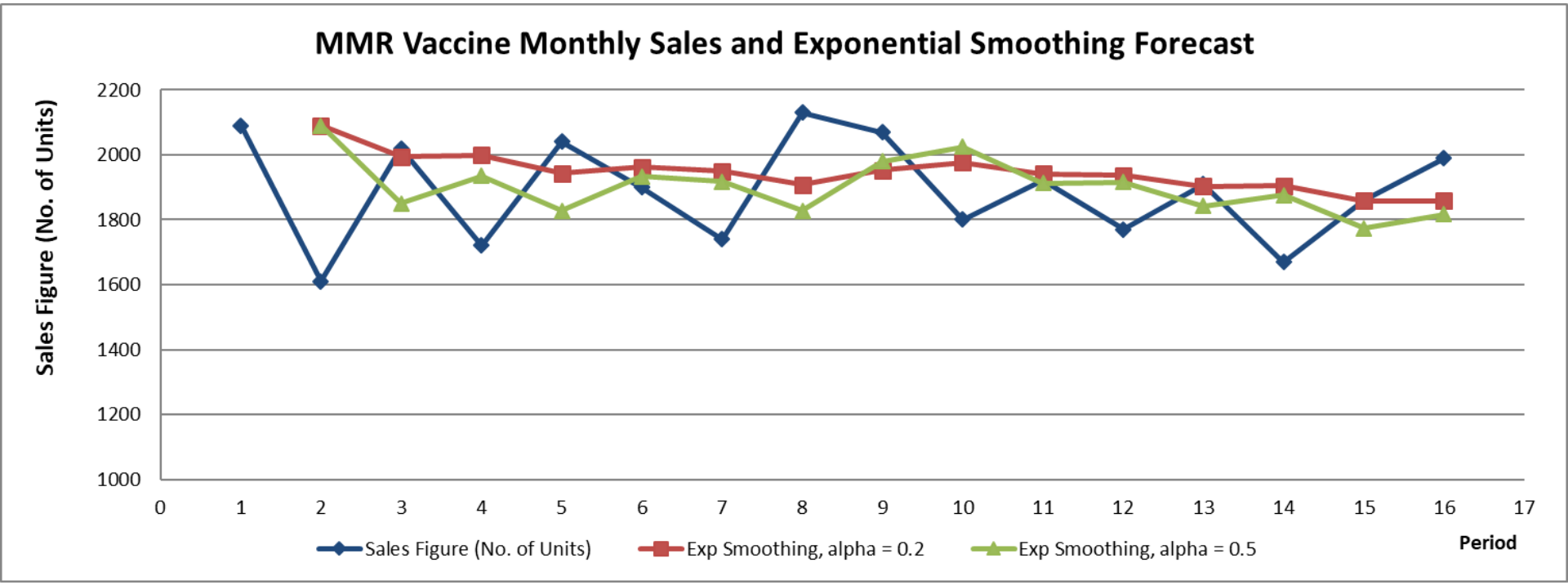
Period	Sales Figure (No. of Units)	**Exp Smoothing, alpha = 0.2**	**Exp Smoothing, alpha = 0.5**
1	2090	#N/A	#N/A
2	1610	2090.00	2090.00
3	2020	1994.00	1850.00
4	1720	1999.20	1935.00
5	2040	1943.36	1827.50
6	1900	1962.69	1933.75
7	1740	1950.15	1916.88
8	2130	1908.12	1828.44
9	2070	1952.50	1979.22
10	1800	1976.00	2024.61
11	1920	1940.80	1912.30
12	1770	1936.64	1916.15
13	1910	1903.31	1843.08
14	1670	1904.65	1876.54
15	1860	1857.72	1773.27
16	1990	1858.17	1816.63

$F_3 = 0.2*1610+(1-0.2)*2090$

Recall:

$F_t = \alpha A_{t-1} + (1 - \alpha) F_{t-1}$

MMR Vaccine Monthly Sales: Simple Exponential Smoothing Solution



For the smoothing constant α,

- When α = 0.2 (compared with 0.5), response to change is slower, with resultant smooth estimators.
- When α = 0.5 (compared with 0.2), response to change is faster, with resultant large variability in the output.

MMR Vaccine Monthly Sales: Forecasting Errors for Moving Average and Exponential Smoothing Methods

Forecasting methods	Measures of forecasting error	
	MAD	MAPE
MA (3-Period)	150.00	7.97%
MA (5-Period)	121.82	6.49%
Weighted MA (3-Period)	151.88	8.08%
Exp Smoothing, (alpha = 0.2)	148.86	8.33%
Exp Smoothing, (alpha = 0.5)	172.83	9.47%

- For the MMR Vaccine Monthly Sales data, Moving Average (5-period) provides the most accurate forecasts from June 2022 to current as its forecasting error is the smallest.
- We can use Moving Average (5-period) to do forecast for the MMR Vaccine Monthly Sales in Oct-23:

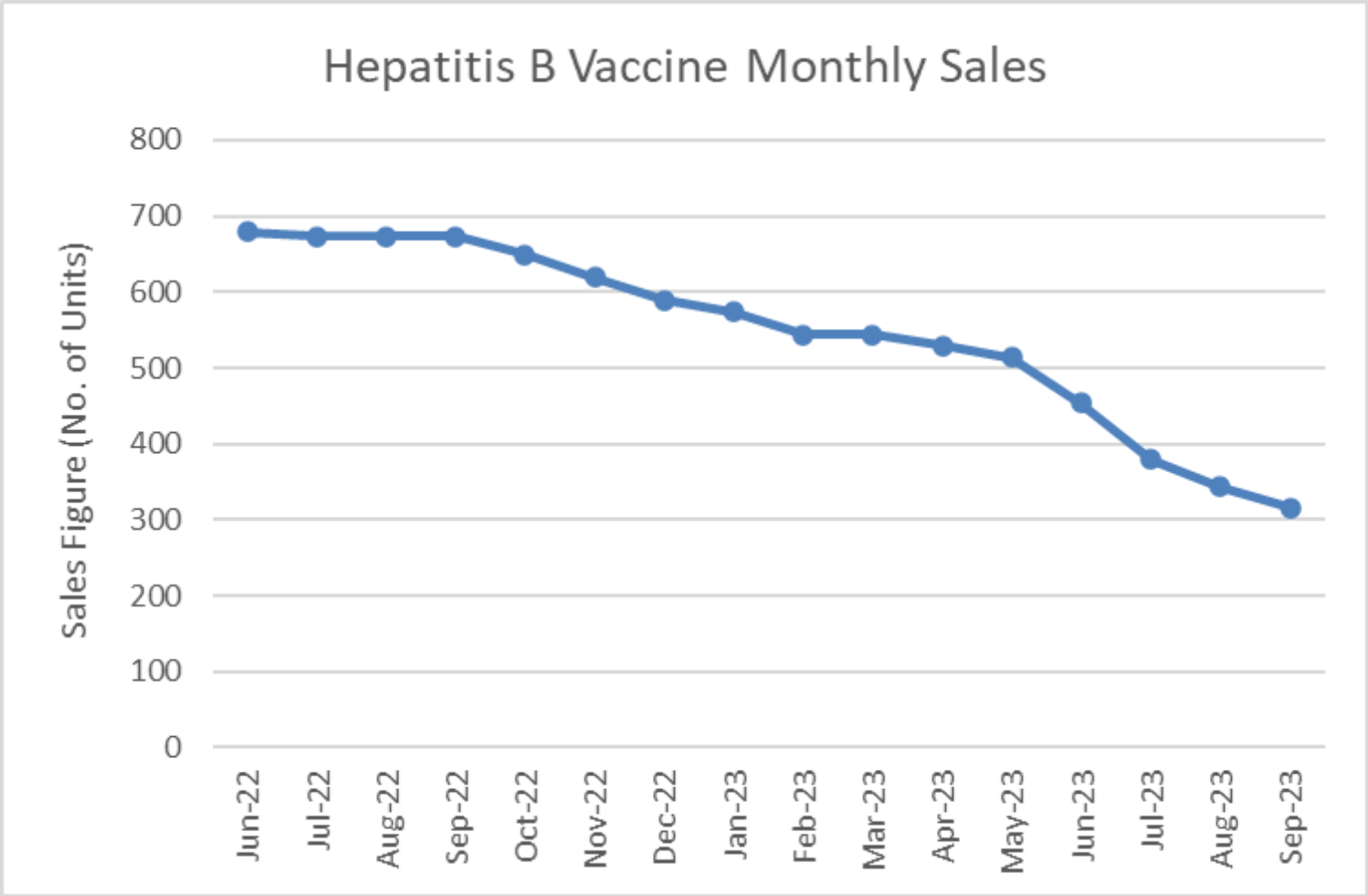
$$F_{17} = \frac{1990+1860+1670+1910+1770}{5}$$

$$=1840.00$$

Period	Sales Figure (No. of Units)	MA, 3-Period	MA, 5-Period
1	2090		
2	1610		
3	2020		
4	1720	1906.67	
5	2040	1783.33	
6	1900	1926.67	1896.00
7	1740	1886.67	1858.00
8	2130	1893.33	1884.00
9	2070	1923.33	1906.00
10	1800	1980.00	1976.00
11	1920	2000.00	1928.00
12	1770	1930.00	1932.00
13	1910	1830.00	1938.00
14	1670	1866.67	1894.00
15	1860	1783.33	1814.00
16	1990	1813.33	1826.00
		1840.00	1840.00

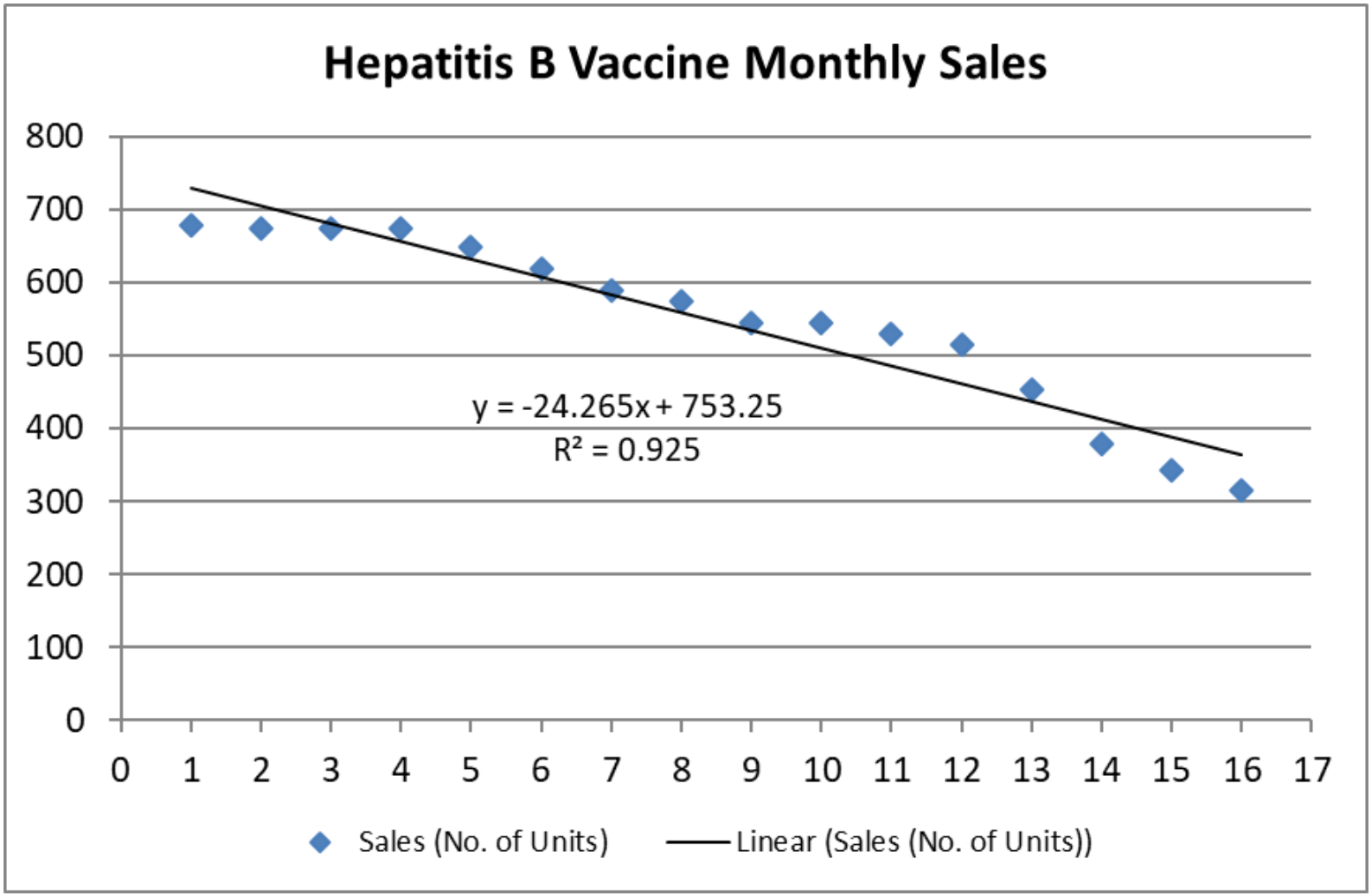
Forecast of Hepatitis B Vaccine Monthly Sales

Hepatitis B Vaccine Monthly Sales



The Hepatitis B Vaccine Monthly Sales data exhibits a downward trend.

Trend Analysis: Forecasting for Hepatitis B Vaccine Monthly Sales



There is a visible downward trend in the Hepatitis B Vaccine Monthly Sales data, will moving average, simple exponential smoothing and Linear Regression methods be suitable to do forecast for this set of data?

Hepatitis B Vaccine Monthly Sales: Forecasting using Holt's Method

- To begin the forecasting process for Holt's Method, Andy can use the results from regression analysis (intercept and slope) as the estimate for L_0 (= 753.25) and T_0 (= -24.2647).
- The choice of smoothing constants plays significant roles in determining accuracy of forecasts.
- Can select different sets of smoothing constants and see which set is better.

E.g. choose α (Alpha) = 0.3 and β (Beta) = 0.6.

Hepatitis B Vaccine Monthly Sales: Forecasting using Holt's Method (α = 0.3, β = 0.6)

Month	Period	A_t (Sales Figure)	L_t	T_t	Holt's H_t
			753.250	-24.265	
Jun-22	1	679	713.9897	-33.2621	728.9853
Jul-22	2	673	678.4094	-34.6530	680.7276
Aug-22	3	673	652.5294	-29.3892	643.7563
Sep-22	4	673	638.0982	-20.4144	623.1402
Oct-22	5	649	627.0786	-14.7775	617.6838
Nov-22	6	619	614.3108	-13.5717	612.3011
Dec-22	7	589	597.2174	-15.6847	600.7391
Jan-23	8	574	579.2728	-17.0406	581.5326
Feb-23	9	544	556.7626	-20.3224	562.2322
Mar-23	10	544	538.7081	-18.9616	536.4402
Apr-23	11	529	522.5225	-17.2960	519.7465
May-23	12	514	507.8586	-15.7168	505.2265
Jun-23	13	454	480.6993	-22.5823	492.1418
Jul-23	14	379	434.3819	-36.8234	458.1170
Aug-23	15	343	381.1910	-46.6439	397.5585
Sep-23	16	316	328.9830	-49.9824	334.5471
Oct-23	17	???			279.0006
Nov-23	18	???			229.0182
Dec-23	19	???			179.0359

T_0: Slope of linear trend line

L_0: intercept of linear trend line

With α = 0.3, β = 0.6

Oct-23, F_{17} = 279.00

Nov-23, F_{18} = 229.02

Dec-23, F_{19} = 179.04

$H_{17} = L_{16} + T_{16}$

$H_{19} = L_{16} + 3 * T_{16}$

$L_{12} = 0.3*A_{12} + (1-0.3)*(L_{11}+T_{11})$

$T_{12} = 0.6*(L_{12}-L_{11}) + (1-0.6)*T_{11}$

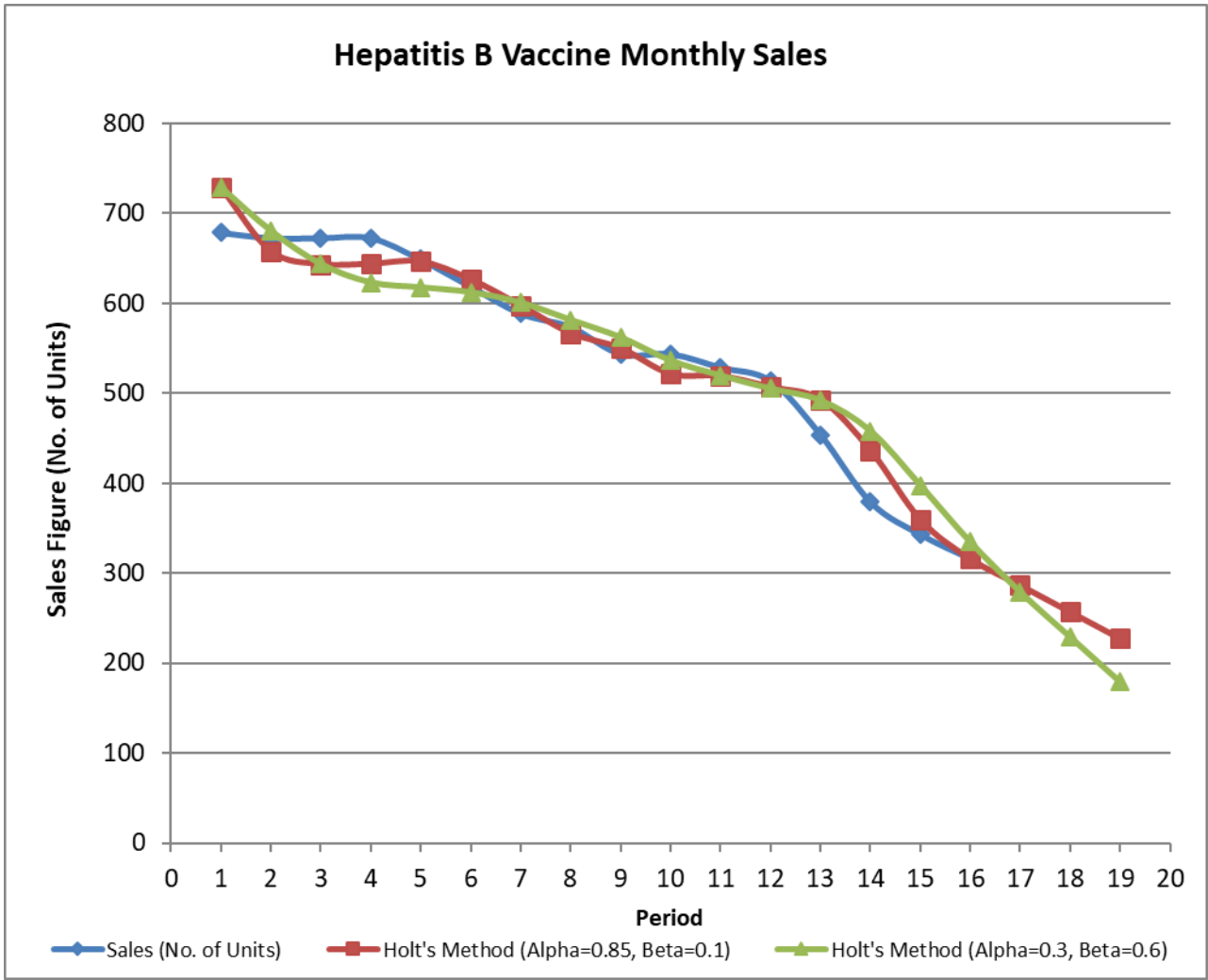
Hepatitis B Vaccine Monthly Sales: Forecasting using Holt's Method

- Comparison of different sets of smoothing constants:

  Set 1: $\alpha = 0.3$ and $\beta = 0.6$; Set 2: $\alpha = 0.85$ and $\beta = 0.10$.

The red line (Set 2) can better capture the change in the historical data than the green line (Set 1) which visibly lags behind the change in the data.

Choice of smoothing constants can significantly affect the performance of the forecasting method.



Hepatitis B Vaccine Monthly Sales: Which Forecasting Method is Better?

Forecasting methods	Measures of forecasting error	
	MAD	MAPE
Linear Regression	27.38	5.71%
Holt's (α = 0.3, β = 0.6)	26.77	5.55%
Holt's (α = 0.85, β = 0.1)	19.24	3.73%

The Holt's Method with α = 0.85 and β = 0.10 gives the best forecast in terms of the forecasting errors.

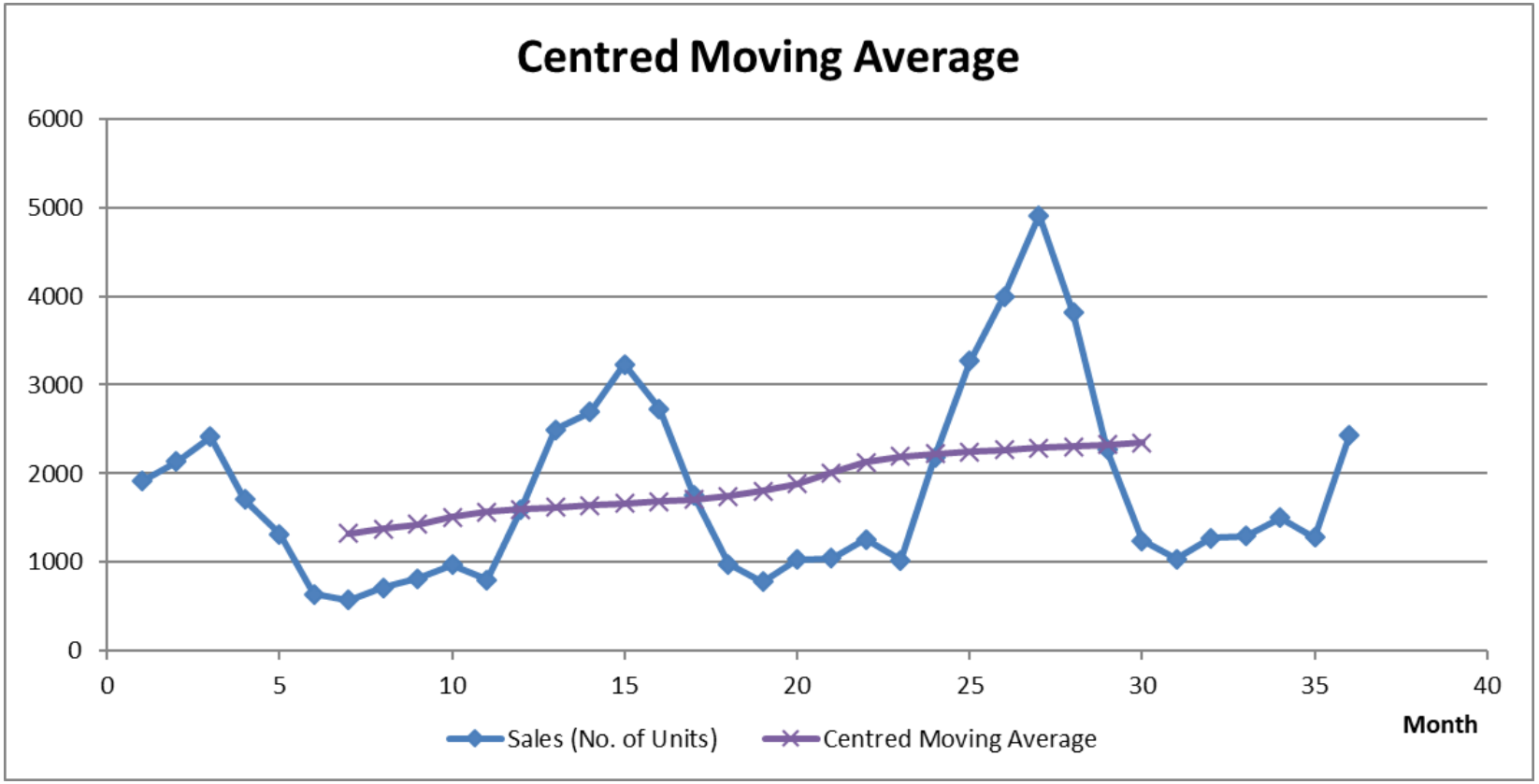
- The Moving Average and Simple Exponential Smoothing methods would not work well because the forecasted values tend to lag when there is a trend in the data set.
  - N-period Moving Average method will lag by (N+1)/2 periods.
  - Simple Exponential Smoothing method with smoothing constant α will lag by 1/α periods.
- The better performance of Holt's Method over Linear Regression is due to:
  - The rate of decrease in the historical data (Hepatitis B Vaccine Monthly Sales) is not constant over the time periods.
  - Linear Regression treats all data in the time series equally.
  - Holt's Method can put more weight on more recent data and therefore can better capture the change in the data.

Forecast of Flu Vaccine Monthly Sales

Flu Vaccine Monthly Sales: CMA and Seasonal Ratio

- By observing the Flu Vaccine Monthly Sales data, the repetitive cycle is 12 months from October to September of the next year. So the number of averaging periods N should be 12.
- As N = 12 is an even number, CMA of Period 7 (Apr-21) will be:
  - (1915+2136+ … +804+1596)/12 = 1300.417 (MA for Period 6.5 <u>put in Period 6</u>)
  - (2136+2417+ … +1596+2496)/12 = 1348.833 (MA for Period 7.5 <u>put in Period 7</u>)
  - (1300.417 + 1348.833)/2 = 1324.625 (CMA for Period 7)
- Because the actual sales of Period 7 (Apr-21) is 574, its Seasonal Ratio will be:
  - 574 / 1324.625 = 0.433

Flu Vaccine Monthly Sales: CMA



- CMA removes seasonal variation from the actual sales data.

Flu Vaccine Monthly Sales: Seasonal Index

- Group the Seasonal Ratio for like seasons and get the average for each season, for example:
  - Season 6 (Mar): (0.561+0.529)/2 = 0.545
  - Season 10 (Jul): (0.642+0.592)/2 = 0.617
- Calculate the normalizing factor:
  - 12/(1.497+1.702+ … +0.991) = 12/12.032 = 0.9973
  - In this case, the normalizing factor is very close to 1
- Multiply the normalizing factor with the average Seasonal Ratio to obtain the Normalized Seasonal Index:

Season	1 (Oct)	2 (Nov)	3 (Dec)	4 (Jan)	5 (Feb)	6 (Mar)	7 (Apr)	8 (May)	9 (Jun)	10 (Jul)	11 (Aug)	12 (Sep)
Normalized Average	1.493	1.697	2.038	1.630	0.998	0.544	0.433	0.531	0.544	0.615	0.489	0.988

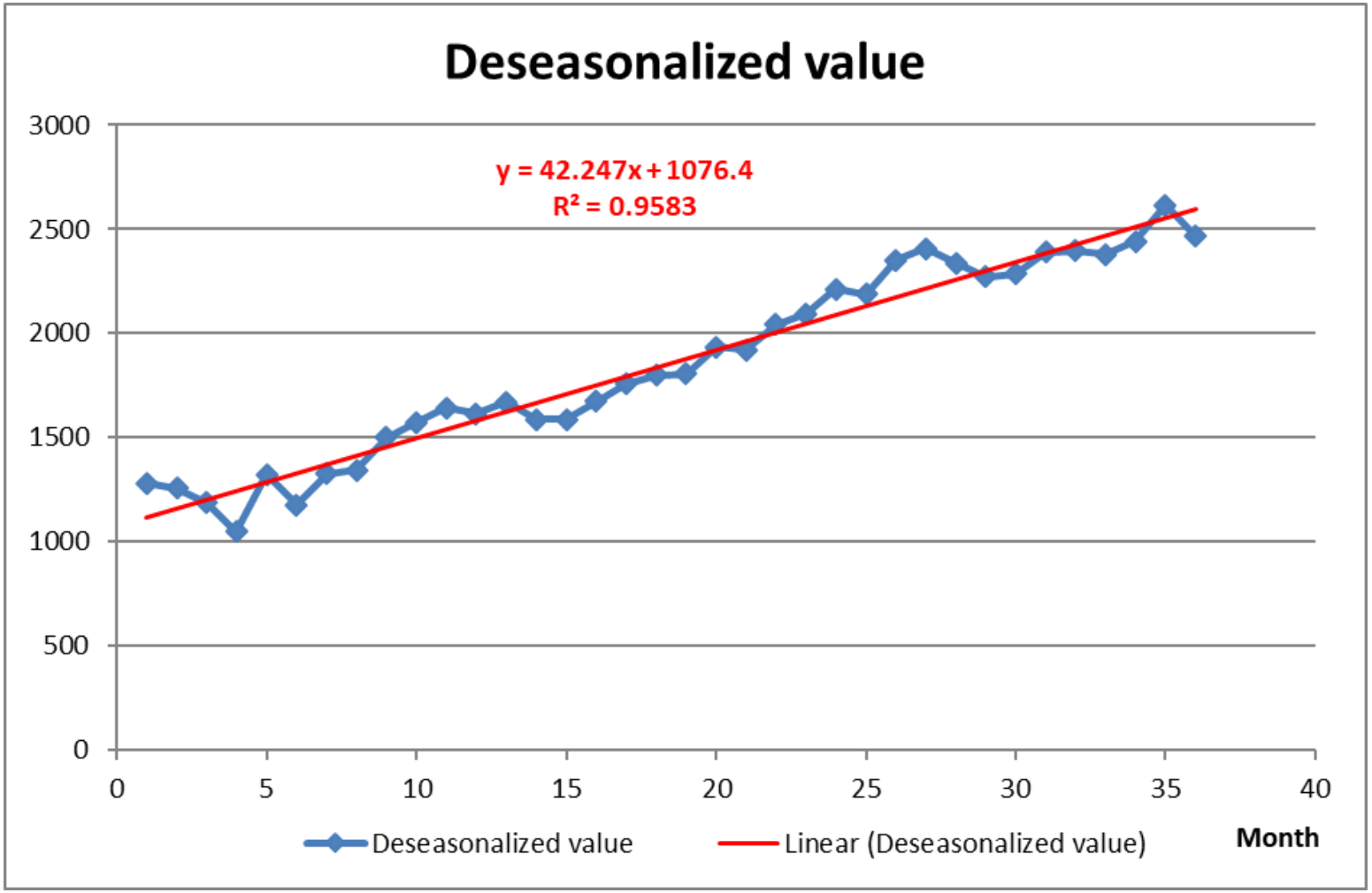
Flu Vaccine Monthly Sales: Deseasonalized Values

Input Data			Seasonal Index Computation				
Month	Sales Figure (No. of Units) (Y)	Time Period (X)	Moving Average (e.g. 12 months)	Centred Moving Average (=(MA_{i-1} + MA_i)/2)	Seasonal Ratio (=Y_i/CMA_i)	Seasonal Index (= normalized average of seasonal ratio)	Deseasonalized value (= Y_i / Seasonal $Index_i$)
Oct-20	1915	1				1.493	1282.413
Nov-20	2136	2				1.697	1258.335
Dec-20	2417	3				2.038	1186.084
Jan-21	1709	4				1.630	1048.382
Feb-21	1319	5				0.998	1321.954
Mar-21	639	6	1300.417			0.544	1175.548
Apr-21	574	7	1348.833	1324.625	0.433	0.433	1326.948
May-21	713	8	1395.500	1372.167	0.520	0.531	1343.987
Jun-21	816	9	1463.333	1429.417	0.571	0.544	1499.887
Jul-21	967	10	1548.167	1505.750	0.642	0.615	1571.494
Aug-21	804	11	1584.500	1566.333	0.513	0.489	1643.201
Sep-21	1596	12	1612.667	1598.583	0.998	0.988	1615.028

* Deseasonalized value = Y_i / Seasonal $Index_i$

$$\text{Deseasonalised value} = \frac{2417}{2.038}$$

Flu Vaccine Monthly Sales: Deseasonalized Value



After removing the seasonal effect, the data exhibits an observable linear trend with random variations.

Flu Vaccine Monthly Sales: Deseasonalized Forecast and Seasonalized Forecast

Step 2: Calculate seasonalized forecast:
Seasonal Index$_i$ * Deseasonalized Forecast$_i$

Time Period (X)	Seasonal Index (= normalized average of seasonal ratio)	Deseasonalized value (= Y_i / Seasonal Index$_i$)	Deseasonalized forecast (=slope*X_i + intercept)	Seasonalized forecast (= Seasonal Index$_i$ *Deseasonalized
1	1.493	1282.413	1118.658	1670.467
2	1.697	1258.335	1160.905	1970.614
3	2.038	1186.084	1203.151	2451.781
4	1.630	1048.382	1245.398	2030.161
5	0.998	1321.954	1287.645	1284.768
6	0.544	1175.548	1329.891	722.898
7	0.433	1326.948	1372.138	593.548
8	0.531	1343.987	1414.385	750.347
9	0.544	1499.887	1456.632	792.467
10	0.615	1571.494	1498.878	922.317
11	0.489	1643.201	1541.125	754.055
12	0.988	1615.028	1583.372	1564.717

Intercept	**1076.411**
Slope	**42.247**
Equation Y(X)	**1076.411+42.247*X**

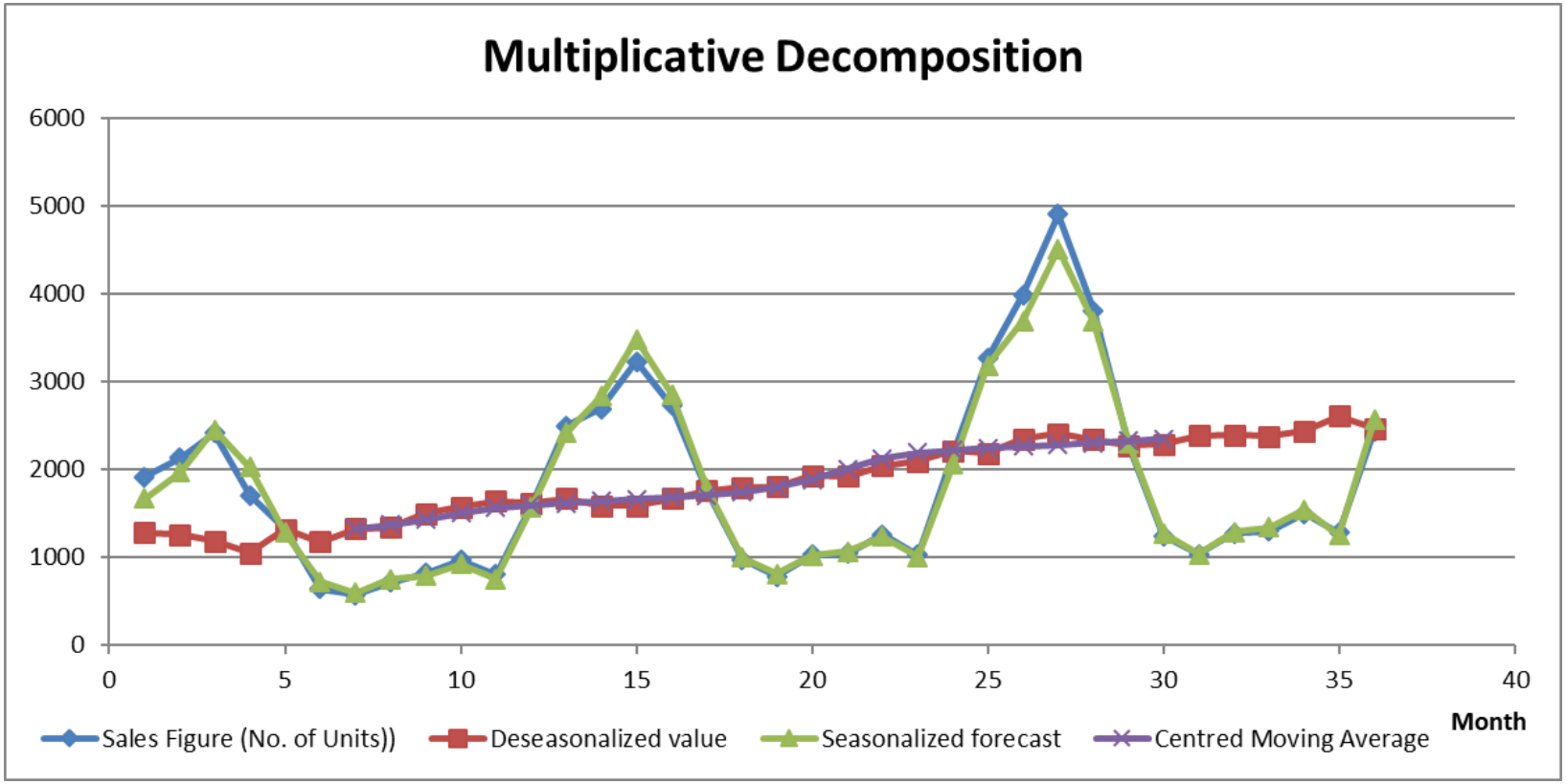
$$2.038 * 1203.151$$

Seasonalized Forecast

Step 1: Calculate deseasonalized forecast:
slope*Xi + intercept

$$\text{Deseasonalized Forecast} = 1076.411 + 42.247 * 3$$

Flu Vaccine Monthly Sales: Actual Value, Deseasonalized Value and Seasonalized Forecast



The seasonalized forecast and the actual values are quite close, so the multiplicative model built is valid to be used to do the forecasting.

Flu Vaccine Monthly Sales: Cyclical and Random Components

Input Data			Forecast & Error	Further decomposition for Cyclicity and Irregular Variations		
Month	Sales Figure (No. of Units) (Y)	Time Period (X)	Seasonalized forecast (= Seasonal Index$_i$ *Deseasonalized forecast$_i$)	Cyclicity*Irregular (=Y$_i$/Seasonalized forecast$_i$)	Cyclicity (3-month MA)	Irregular Variations (=Cyclicity*Irregular$_i$/Cyclicity$_i$)
Oct-20	1915	1	1670.467	1.146		
Nov-20	2136	2	1970.614	1.084	1.072	1.011
Dec-20	2417	3	2451.781	0.986	0.971	1.016
Jan-21	1709	4	2030.161	0.842	0.951	0.885
Feb-21	1319	5	1284.768	1.027	0.917	1.119
Mar-21	639	6	722.898	0.884	0.959	0.922
Apr-21	574	7	593.548	0.967	0.934	1.036
May-21	713	8	750.347	0.950	0.982	0.967
Jun-21	816	9	792.467	1.030	1.009	1.020
Jul-21	967	10	922.317	1.048	1.048	1.000
Aug-21	804	11	754.055	1.066	1.045	1.020
Sep-21	1596	12	1564.717	1.020	1.038	0.983

Step 3: Estimate the Irregular Variations

Irregular Variations = (Cyclicity * Irregular$_i$) / Cyclicity$_i$

$$I = \frac{0.967}{0.934}$$

$$C * I = \frac{574}{593.548}$$

$$MA = \frac{0.884 + 0.967 + 0.950}{3}$$

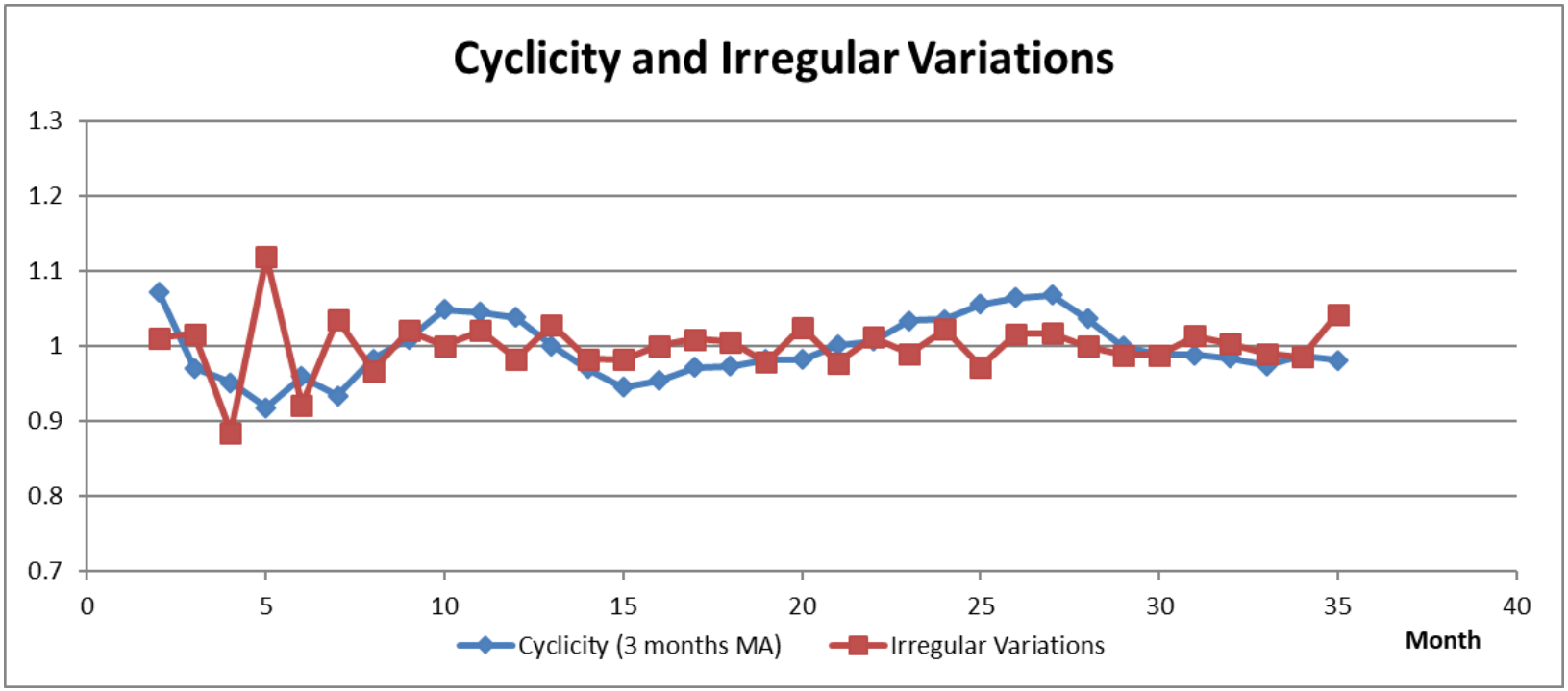
Step 1: Estimate the Cyclicity*Irregular

Cyclicity * Irregular$_i$ = Y$_i$ / Seasonalized forecast$_i$

Step 2: Estimate the Cyclicity factor

Use 3-month MA to remove the Random factor to estimate the Cyclicity factor.

Flu Vaccine Monthly Sales: Cyclical and Random Components

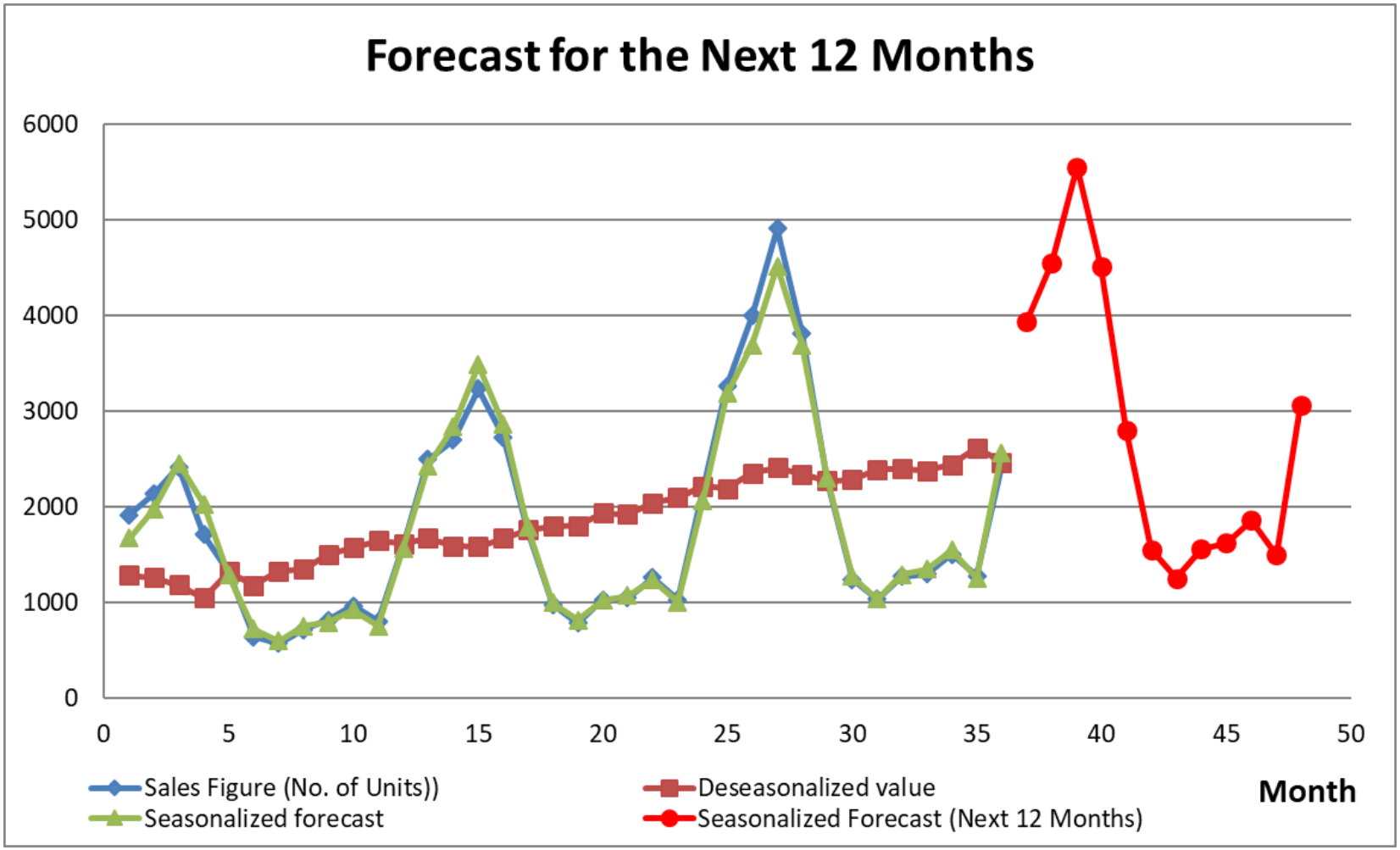


Most of the cyclical and random factors are between 0.9 and 1.1, which can be considered as minor. Therefore, they can be ignored for the forecast of the Flu Vaccine Monthly Sales in the next 12 months.

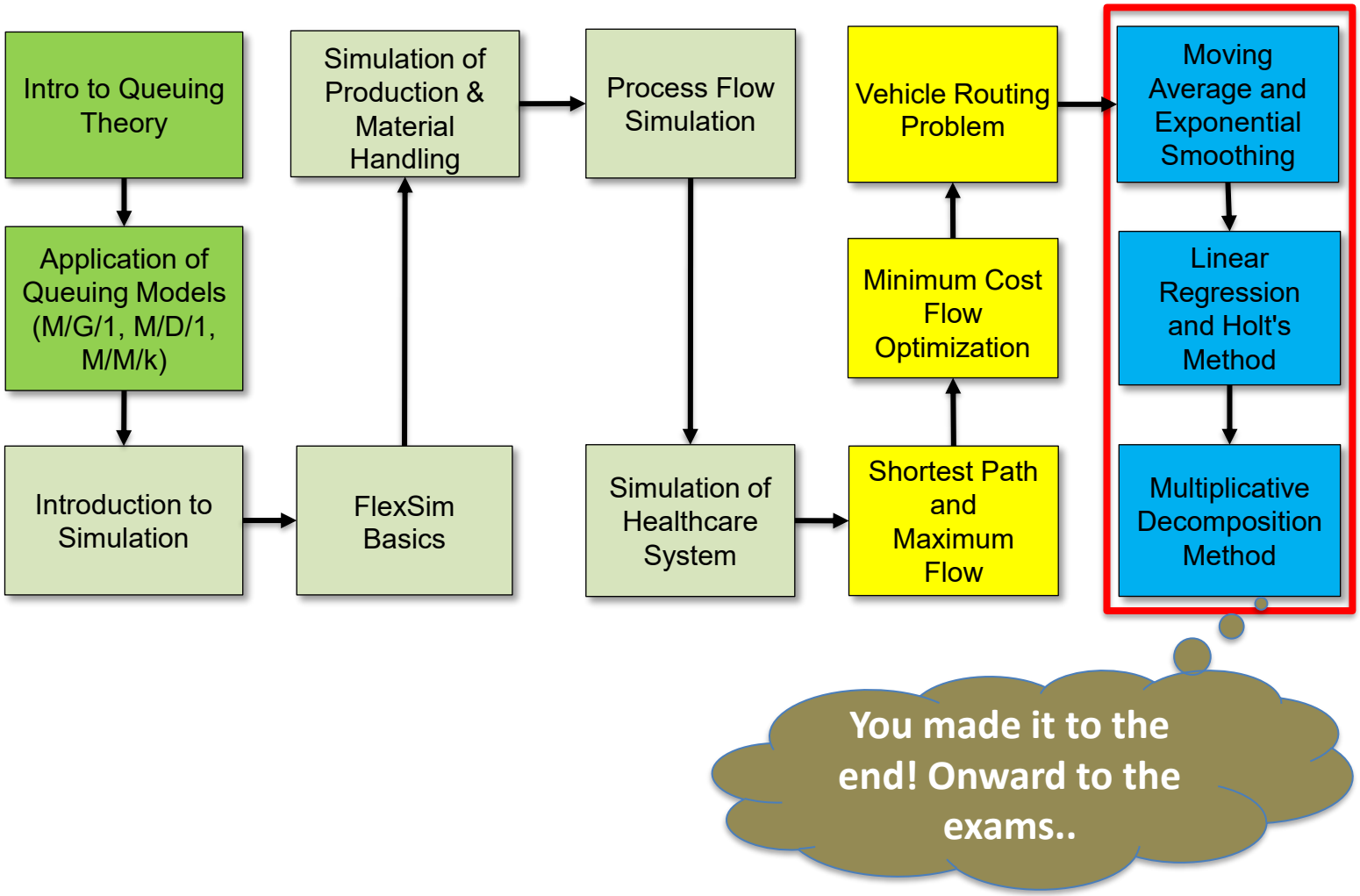
Flu Vaccine Monthly Sales: Forecast for the Next 12 Months using the Model Built

Forecast for the next 12 months				
Month	Time Period	Deseasonalized Forecast	Seasonal Index	Seasonalized Forecast (Next 12 Months)
Oct-23	37	2639.540	1.493	3941.567
Nov-23	38	2681.786	1.697	4552.282
Dec-23	39	2724.033	2.038	5551.032
Jan-24	40	2766.280	1.630	4509.397
Feb-24	41	2808.526	0.998	2802.252
Mar-24	42	2850.773	0.544	1549.613
Apr-24	43	2893.020	0.433	1251.438
May-24	44	2935.267	0.531	1557.192
Jun-24	45	2977.513	0.544	1619.889
Jul-24	46	3019.760	0.615	1858.173
Aug-24	47	3062.007	0.489	1498.205
Sep-24	48	3104.253	0.988	3067.680

Flu Vaccine Monthly Sales: Seasonalized Forecast for the Next 12 Months



Summary

- Multiplicative Decomposition is an intuitive method for forecasting, which is useful for time series analysis with fixed describing parameters.
- With Multiplicative Decomposition method, forecasting can be done for a time series with the Seasonal and the Trend components.

E211 Progression